

AUTOMATED INTERPRETABILITY METRICS DO NOT DISTINGUISH TRAINED AND RANDOM TRANSFORMERS

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ABSTRACT

Sparse autoencoders (SAEs) are widely used to extract sparse, interpretable latents from transformer activations. We ask a direct, quantitative question: do commonly used SAE quality metrics and automatic explanation pipelines actually separate a trained transformer from a randomly initialized one (e.g., where parameters are sampled i.i.d. from a Gaussian)? To answer it, we run a controlled head-to-head comparison across the full Pythia suite (70M to 6.9B parameters) and four distinct randomization schemes, holding the SAE architecture, training data (RedPajama), and evaluation pipeline fixed so that any gap is attributable to the underlying model rather than the measurement setup. The central empirical finding is a striking convergence: on aggregate auto-interpretability scores (fuzzing AUROC) and standard reconstruction metrics, the randomized variants track the trained model closely and sit far from a chance-level control, so the trained-vs-random margin that these metrics are supposed to reveal is largely absent. This means a high aggregate auto-interpretability score, on its own, is not evidence that learned, computationally relevant features have been recovered—the same score is reached by weights that never learned. We therefore recommend treating common SAE metrics as useful but insufficient proxies for mechanistic interpretability, and we show that a targeted measure of feature ‘abstractness’ (token-distribution entropy) recovers a trained-vs-random distinction that the aggregate scores miss, motivating routine randomized baselines.

1 INTRODUCTION

Sparse autoencoders (SAEs) are a popular tool in mechanistic interpretability research, with the aim of disentangling the internal representations of neural networks by learning sparse, interpretable features from network activations (Elhage et al., 2022; Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). An autoencoder with a high-dimensional hidden layer is trained to reconstruct activations while enforcing sparsity (Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024), with the aim of discovering the underlying concepts or ‘features’ learned by the network (Park et al., 2023; Wattenberg and Viégas, 2024). Developing better SAEs relies on quantitative evaluation metrics like auto-interpretability scores that measure agreement between generated explanations and activation patterns (Bills et al., 2023; Paulo et al., 2024; Karvonen et al., 2024a).

For an interpretability method to be considered robust, its evaluation metrics should distinguish features learned through training from artifacts arising from the data or model architecture. The cleanest way to test this is a direct comparison: run the method on a trained model and on a strong null model with randomly initialized weights, holding everything else fixed, and measure the gap (Adebayo et al., 2020). We apply exactly this sanity check to SAEs, and the comparison is telling—across most of the settings we study, the trained-vs-random gap that common quantitative metrics should reveal is small or absent. Concretely, SAEs trained on transformers with random parameters yield latents whose auto-interpretability scores (Bills et al., 2023; Paulo et al., 2024) closely match those from a fully trained model, while a chance-level control sits far below both—so the metric separates trained from control but fails to separate trained from random.

This near-parity between trained and random models directly constrains what a high SAE-quality score can be taken to mean. Because the same aggregate auto-interpretability score is attainable without any training, a high score alone cannot certify that an SAE has identified complex, learned

computations. Rather, a large fraction of what these scores measure appears to reflect simpler statistical properties of the training data (Dooms and Wilhelm, 2024) or architectural inductive biases that are present even at random initialization. Indeed, one could argue that a randomly initialized network still performs a basic form of computation, such as preserving or amplifying the sparse structure of its inputs (Section 4). From this perspective, the SAE is faithfully interpreting this simple, inherent computation—which is precisely why the trained and random scores land so close together.

While some SAE features from trained models clearly arise from learned computation, the small trained-vs-random gap we measure on aggregate metrics shows that these metrics, by themselves, cannot decide whether a given SAE has captured such features. These quantitative results have important implications for mechanistic interpretability research. In particular, they argue for metrics with demonstrable discriminative power between artifacts and genuinely learned computations, and for validating interpretability techniques against randomized null models whose scores can be compared directly against the trained model’s.

Finally, we speculate about why these patterns might emerge. At a high level, there are two hypotheses: (1) the input data already exhibits superposition, and randomly initialized neural networks largely preserve this superposition; and (2) randomly initialized neural networks amplify or even introduce superposed structure to the input data (e.g., given dense input generated i.i.d. from a Gaussian). We present toy models to demonstrate the plausibility of these hypotheses in Section 4 but defer conclusions as to the mechanism responsible to future work.

2 RELATED WORK

Sparse dictionary learning Under a different name, ‘superposition’ in visual data is one of the foundational observations of computational neuroscience. Olshausen and Field (1996; 1997) showed that the receptive fields of simple cells in the mammalian visual cortex can be explained as a result of sparse coding, i.e., representing a relatively large number of signals (sensory information) by simultaneously activating a small number of elements (neurons). Coding theory offers a perspective on efforts to extract the ‘underlying signals’ responsible for neural network activations (Marshall and Kirchner, 2024).

Sparse dictionary learning (SDL) approximates a set of input vectors by linear combinations of a relatively small number of learned basis vectors. The learned basis is usually overcomplete: it has a greater dimension than the inputs. SDL algorithms include Independent Component Analysis (ICA), which finds a linear representation of the data such that the components are maximally statistically independent (Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000). Sparse autoencoders (SAEs) are a simple neural network approach (Lee et al., 2006; Ng, 2011; Makhzani and Frey, 2014). Typically, an autoencoder with a single hidden layer that is many times larger than the input activation vectors is trained with an objective that imposes or incentivizes sparsity in its hidden layer activations to try to find this structure. A **latent** is a single neuron (dimension) in the autoencoder’s hidden layer.

Mechanistic interpretability Recently, it has become common to understand ‘features’ or concepts in language models as low-dimensional subspaces of internal model activations (Park et al., 2023; Wattenberg and Viégas, 2024; Engels et al., 2024). If such sparse or ‘superposed’ structure exists, we expect to be able to ‘intervene on’ or ‘steer’ the activations, i.e., to modify or replace them to express different concepts and so influence model behavior (Meng et al., 2022; Zhang and Nanda, 2023; Heimersheim and Nanda, 2024; Makelov, 2024; O’Brien et al., 2024).

SAEs are a popular approach for discovering features, where one typically trains a single autoencoder to reconstruct the activations of a single neural network layer, e.g., the transformer residual stream (Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). Many SAE architectures have been suggested, which commonly vary the activation function applied after the linear encoder (Makhzani and Frey, 2014; Gao et al., 2024; Rajamanoharan et al., 2024b; Lieberum et al., 2024). SAEs have also been trained with different objectives (Braun et al., 2024; Farnik et al., 2025) and applied to multiple layers simultaneously (Yun et al., 2021; Lawson et al., 2024; Lindsey et al., 2024).

Besides reconstruction errors and preservation of the underlying model’s performance, SAEs have been evaluated according to whether they capture specific concepts (Gurnee et al., 2023; Gao et al.,

2024) or factual knowledge (Huang et al., 2024; Chaudhary and Geiger, 2024), and whether these can be used to ‘unlearn’ concepts (Karvonen et al., 2024b).

Automatic neuron description SAEs often learn tens of thousands of latents, which are infeasible to describe by hand. Yun et al. (2021) find the tokens that maximally activate a dictionary element from a text dataset and manually inspect activation patterns. Instead, researchers typically collect latent activation patterns over a text dataset and prompt a large language model to explain them (e.g. Bills et al., 2023; Foote et al., 2023). These methods have been widely adopted (e.g. Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024).

Bills et al. (2023) generate an explanation for the activation patterns of a language-model neuron over examples from a dataset, simulate the patterns based on the explanation, and score the explanation by comparing the observed and simulated activations. This method is commonly known as auto-interpretability (as in self-interpreting). Paulo et al. (2024) introduce classification-based measures of the fidelity of automatic descriptions that are inexpensive to compute relative to simulating activation patterns and an open-source pipeline to compute these measures. Choi et al. (2024) use best-of- k sampling to generate multiple explanations based on different subsets of the examples that maximally activate a neuron. Importantly, they fine-tune Llama-3.1-8B-Instruct on the top-scoring explanations to obtain inexpensive ‘explainer’ and ‘simulator’ models.

Polysemanticity Lecomte et al. (2024) noted that neurons may become polysemantic incidentally. A polysemantic neuron (basis dimension) of a network layer represents multiple interpretable concepts (Elhage et al., 2022; Scherlis et al., 2023); unsurprisingly, individual neurons in a randomly initialized network may be polysemantic. By contrast, our work studies *superposition* (Elhage et al., 2022; Chan, 2024), which pertains to the representations learned across a whole network layer as opposed to any individual neuron. In particular, superposition allows a network layer as a whole to represent a larger number of (sparse) features than the layer has (dense) neurons by sparse coding (only a few concepts are active at a time, i.e., a given token position).

Training only the embeddings Zhong and Andreas (2024) showed that transformers learn surprising algorithmic capabilities when only the embeddings are trained and no other parameters. These results demonstrate that the behavior of a randomly initialized transformer can be shaped to a surprising extent by training only a few parameters. However, our setting is very different: besides considering SAEs, we randomize *all* the parameters, including the embeddings, in our ‘Step-0’ and ‘Re-randomized incl. embeddings’ variants. Our ‘Re-randomized excl. embeddings’ variant uses pre-trained embeddings, but we do not train those embeddings with fixed, randomized weights. Instead, we freeze the pre-trained embeddings and randomize the other weights (Section 3).

Random transformers for board games Karvonen et al. (2024c) report a clear trained-vs-random gap: SAEs extracted considerably more meaningful structure from chess games with pre-trained transformers than with random weights. This contrast is consistent with ours once the data regime is taken into account, because board-game data differs sharply from language data. In particular, there is reason to expect that language is sparse (e.g., a particular concept such as ‘serendipitous’ appears only rarely), and that this sparse structure is ‘aligned’ with conceptual meaning. In board games this alignment need not hold: a useful concept such as a knight fork does not necessarily turn up sparsely, so the trained model has more to add and the measured gap widens.

Random one-layer transformers Bricken et al. (2023) report that auto-interpretability scores discriminated effectively between random and trained one-layer transformers. Our results refine this picture along the model-size axis: the trained-vs-random gap is sizeable for the smallest models (e.g., Pythia-70m) but shrinks steadily as scale grows, until at Pythia-6.9b it has nearly closed.

3 RESULTS

To make the trained-vs-random comparison as controlled as possible, we trained per-layer SAEs on the residual stream activation vectors of transformer language models from the Pythia suite, spanning two orders of magnitude in scale (70M to 7B parameters) (Biderman et al., 2023). Every SAE shares the same architecture, training data, and evaluation pipeline; the only thing that varies

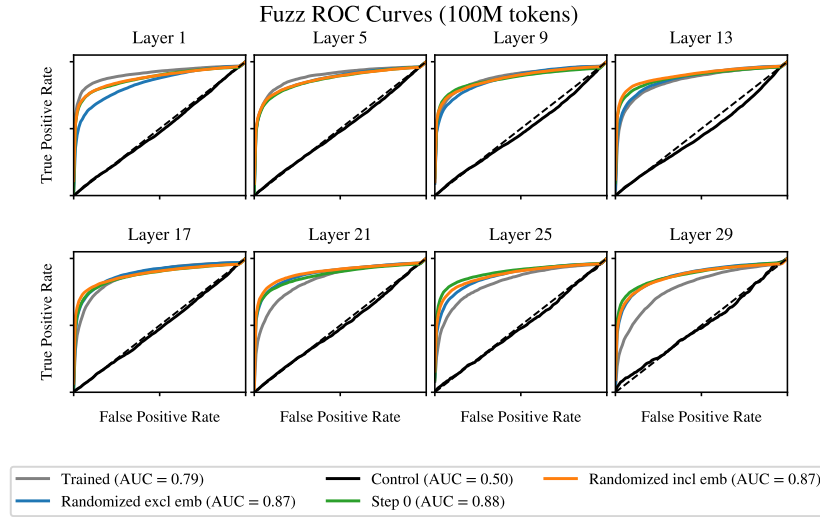


Figure 1: Head-to-head ‘fuzzing’ auto-interpretability at Pythia-6.9b, ROC curve by layer (100 latents sampled per SAE). The comparison is unambiguous: the trained model (gray) and every randomized variant (colored) trace nearly the same ROC curves at every layer, while the chance-level control (black, dotted) sits far below all of them. In other words, the metric cleanly separates real embeddings from the random-embedding control but leaves essentially no margin between trained and randomized weights—so a high aggregate AUROC alone cannot attribute latents to learned computation. See Figure 2 for the same pattern across additional metrics and model sizes, and Appendix E for consistency across five random seeds.

is the underlying transformer, so any difference in the resulting metrics is attributable to how the transformer’s weights were obtained. We compared five variants of that underlying transformer, four of which never underwent (full) training:

- **Trained:** The usual, trained model.
- **Re-randomized incl. embeddings:** All the model parameters, including the embeddings, are re-initialized by sampling Gaussian noise with mean and variance equal to the values for each of the original, trained weight matrices.
- **Re-randomized excl. embeddings:** As above, except the embedding and unembedding weight matrices are not re-initialized, i.e., are the same as the original, trained model.
- **Step-0:** For Pythia models, the step0 revisions are available, which are the original model weights at initialization, i.e., before any learning (Biderman et al., 2023).
- **Control:** The original, trained model, except where the input token embeddings are replaced at inference time by sampling i.i.d. standard Gaussian noise for each token, such that a given token does not have a consistent embedding vector. For this variant, we expect auto-interpretability to perform at the level of chance.

For our primary experiments, every SAE was trained on the same 100M tokens from RedPajama (Weber et al., 2024) with a 10M-token activation buffer, so training data is held constant across variants (Appendix C confirms the comparison is unchanged at 1B tokens). Layer coverage was set by compute budget: for models with fewer than 410M parameters, we trained an SAE at every layer; for Pythia-1b, we trained SAEs at every second layer; and for Pythia-6.9b, we trained SAEs at every fourth layer.

Unless otherwise stated, we trained k -sparse autoencoders (also known as TopK SAEs; Makhzani and Frey 2014; Gao et al. 2024), fixing the expansion factor at $R = 64$ and the sparsity at $k = 32$ so that this choice cannot drive the comparison. The trained-vs-random pattern proves robust to this choice: sweeping Pythia-160m over expansion factors from 16 to 128 (powers of 2) and sparsities of

16 and 32 leaves the comparison intact (Figure 18). The training implementation is based on Belrose et al. (2025); our evaluations are based on Caden et al. (2025) and Karvonen et al. (2024a).

Auto-interpretability Auto-interpretability turns explanation quality into a number we can compare across variants: a feature explanation that names a concept is fed to a classifier that predicts whether the concept appears in the text, and the classifier is scored by a standard metric—the area under the receiver operating characteristic (ROC) curve (AUROC). Paulo et al. (2024) proposed ‘fuzzing’ and ‘detection’ classification tasks to evaluate feature explanations. For ‘fuzzing’ scoring, both positive and negative examples of tokens (i.e., with non-zero and zero activation values, respectively) for a given latent are delimited with special characters, and a language model is prompted to identify which examples have been correctly delimited for the latent given its explanation. For ‘detection’, a language model is asked to identify which examples contain activating tokens for each feature. Bills et al. (2023) originally proposed ‘simulation’ scoring, based on the correlation between predicted and observed activations, but this method is expensive to compute.

Except where noted, we report ‘fuzzing’ AUROC as our headline auto-interpretability metric, because it has been shown to track the more expensive simulation scoring (Paulo et al., 2024); the corresponding ‘detection’ AUROC curves in Appendix B tell the same story. To keep every variant on equal footing, we scored each SAE (i.e., underlying model, variant, and layer) on the same protocol: a random sample of 100 latents per SAE, using the implementation of Paulo et al. (2024). Explanations and predictions were generated with the Meta-Llama-3.1-70B-Instruct-AWQ-INT4 model, which is both larger than the 8B models of Choi et al. (2024) and open-source, unlike Bills et al. (2023).

The comparison yields a clear separation of one variant and a near-collapse of the rest: measured by fuzzing AUROC, the trained and randomized models cluster tightly together, while only the control falls away toward chance (Figures 1 and 2). That the trained and randomized ROC curves are nearly indistinguishable—while both clearly beat the control—is direct evidence that ‘fuzzing’ auto-interpretability applied to SAE latent explanations does not, on its own, resolve trained from randomized transformers.

Evaluation To check whether the near-parity is specific to auto-interpretability, we placed the standard SAE reconstruction metrics side by side with the auto-interpretability AUROC across Pythia models from 70M to 6.9B parameters. The same ordering recurs on every metric: the randomly initialized and re-randomized models (Figure 2; blue, green, orange lines) sit much closer to the trained model (gray lines) than to the control (black lines), so the trained-vs-random gap is small while the everything-vs-control gap is large.

The control is the consistent exception, and its numbers make the contrast quantitative: its cosine similarity between the original activation and the SAE reconstruction, and its explained variance, are markedly lower than for every other variant, and its reconstruction error grows across layers while all other variants improve. For the random control, this can perhaps be explained by the fact that a Gaussian is the highest entropy distribution with fixed mean and variance (Jaynes, 2003); we speculate that Gaussian vectors are the ‘least structured’, in some sense, and thus hardest for SAEs to reconstruct. As Gaussian-distributed activations are propagated through successive layers, we would expect the activations to become less Gaussian and perhaps more ‘sparse’, i.e., easier to reconstruct (Section 4).

A finer comparison among the non-trained variants is also informative: the two re-randomized variants (blue and orange lines) match the trained model more closely than the at-initialization Step-0 variant (green line) does. This gap between the re-randomized and Step-0 variants is clearest in the L^1 norm at larger model sizes. We attribute it to parameter scale: norms can differ substantially between a trained model and its state at initialization, whereas our re-randomization procedure was designed to preserve the per-matrix parameter norms of the trained model. The scale of parameters at different layers may be important, e.g., to control the growth of activations as they progress through the residual stream (Liu et al., 2020). The AUROC panels add a consistent scale trend: for every variant except the control, AUROC rises monotonically with model size. We speculate that features become more specific as SAE size increases: in smaller SAEs, each latent must explain more of the input, making the classification task easier for larger SAEs.

Row five of Figure 2 reports the cross-entropy (CE) loss score, or loss recovered, against model layer: the increase in loss when the original activations are replaced by their SAE reconstructions, divided

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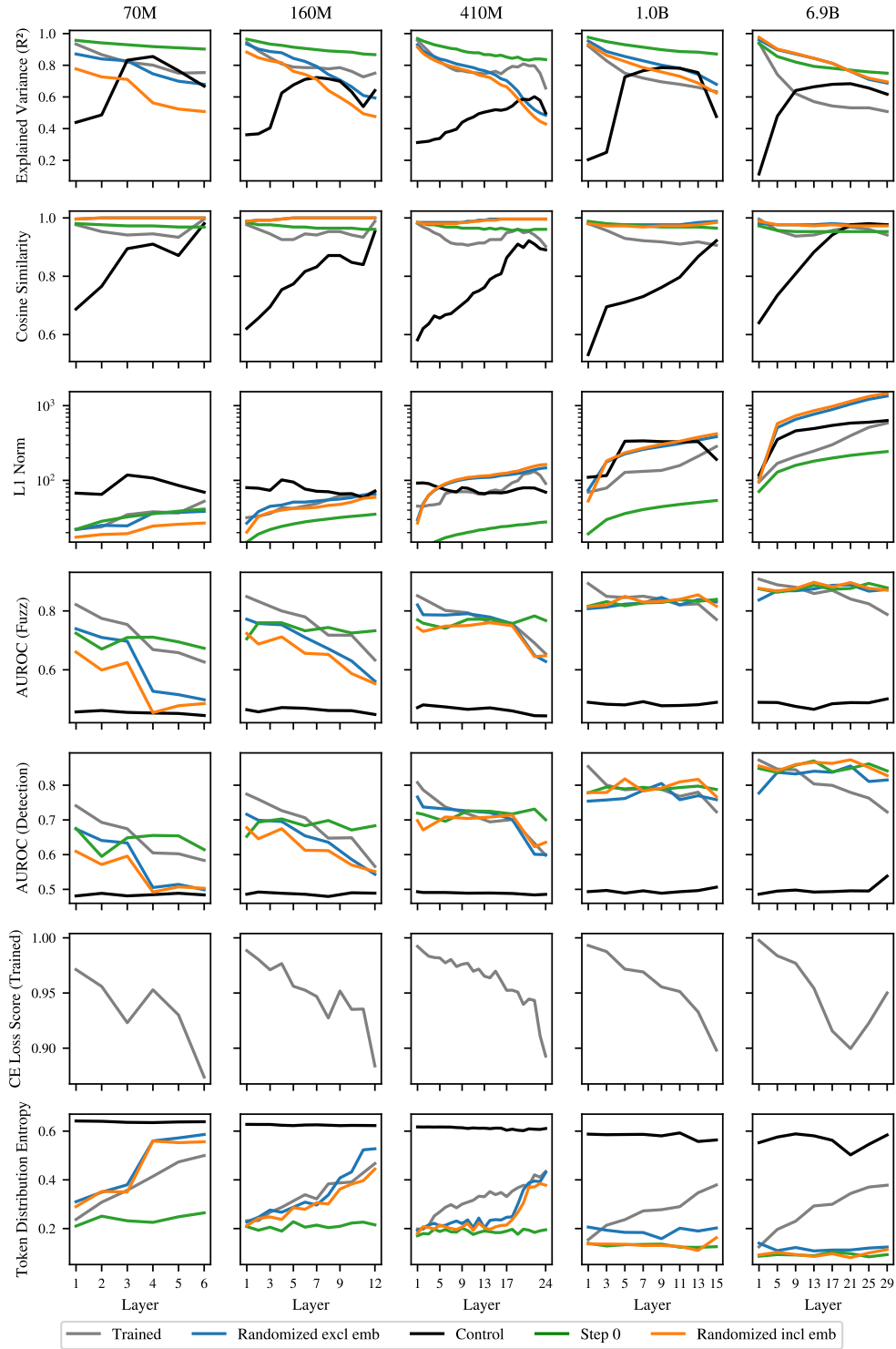


Figure 2: Head-to-head SAE performance across the Pythia suite (70M to 6.9B parameters), one column per model size and one row per metric. The comparison is consistent across every metric and scale: the trained variant and the three randomized variants track one another closely, layer for layer, whereas the control is the lone outlier. The trained-vs-random curves converge further as model size grows. In short, four variants that differ fundamentally in how their weights were obtained—one trained, three never trained—land on comparable performance, and only the random-embedding control separates out.

by the increase when they are replaced by zeros (‘ablated’). As a sanity check on our pipeline, the trained-variant CE loss scores land in the same range as SAEs reported elsewhere in the literature (e.g., Kissane et al., 2024; Rajamanoharan et al., 2024a; Mudide et al., 2024), confirming that our trained SAEs are competitive. This one metric is the exception to the near-parity story, but only because it is not comparable across variants: the CE loss score is meaningful only for the trained variant, since for every randomized variant the underlying loss is very poor whether the original or the reconstructed activations are used.

Latent explanation complexity Given how close the aggregate auto-interpretability and reconstruction scores are, we asked whether any measurable quantity still separates trained from randomized transformers—expecting the two to nonetheless discover qualitatively different features. Specifically, we expected SAEs on the randomized variants to recover only simple features tied to surface characteristics of the input text, and not the more complex, abstract features seen with trained transformers (Templeton et al., 2024). For qualitative examples, we provide a random sample of features and the corresponding maximally activating dataset examples for each variant of Pythia-6.9b in Appendix J, and more detailed information in Appendix L.

Anecdotally, we have observed that a significant proportion of SAE latents have non-zero activations only on a single token or a small number of distinct tokens within a text dataset (e.g., Lin and Bloom, 2024; Dooms and Wilhelm, 2024). Hence, a simple measure of the complexity of an SAE latent given a set of maximally activating examples is the degree to which the latent activates on a single token ID or multiple distinct IDs. Specifically, we quantify the number of token IDs in terms of the entropy of the observed distribution of latent activations over tokens: the greater the entropy, the more ‘spread out’ the latent activations, and the less token-specific the latent. We take this distribution to be the total latent activation per token across the set of maximally activating examples used to generate explanations for auto-interpretability. We show the relationship between entropy and ‘fuzzing’ AUROC score for individual latents in Appendix H.

The last row of Figure 2 reports this token-distribution entropy, and unlike the aggregate scores it does separate the variants into three distinct regimes. The control has consistently high entropy, as expected: because each token’s embedding is resampled i.i.d. from a Gaussian on every occurrence (Section 3), no token has a stable embedding, so activations spread across many tokens. The trained variant shows the opposite trend—its entropy rises with depth, meaning that deeper layers are progressively less likely to concentrate a latent’s maximally activating examples on a single token, exactly the signature of more abstract, less token-tied features. Critically, the randomized variants occupy the low-entropy regime: their entropy sits below both the trained and control variants and stays low across depth, indicating latents that fire specifically at one or a few token IDs. Thus a single, cheap statistic separates trained from randomized transformers along the depth axis precisely where the aggregate auto-interpretability scores could not.

Read together with the near-parity on aggregate scores, this contrast localizes what the standard metrics miss: the ‘abstractness’ of SAE features. Token-distribution entropy is only a proxy for abstractness, but the comparison it enables is clean—the randomized variants match the trained variant on aggregate auto-interpretability precisely because both readily learn simple, single-token features that score well. The two part company only on the depth trend: entropy climbs with layer for the trained variant but stays flat and low for the randomized variants, so only the trained model builds progressively more complex features with depth.

4 A TOY MODEL OF SUPERPOSITION IN RANDOM NETWORKS

Section 1 offered two candidate explanations for why randomized transformers score so close to trained ones: either the input data already exhibits superposition that the network preserves, or the network itself amplifies or introduces superposition. Here we test both with toy models designed so that the relevant quantity—how easily an SAE reaches low reconstruction error at high sparsity—can be measured and compared directly across inputs, outputs, and Gaussian controls. The measurements support both mechanisms to some degree; we leave the question of which dominates in real randomized transformers to future work.

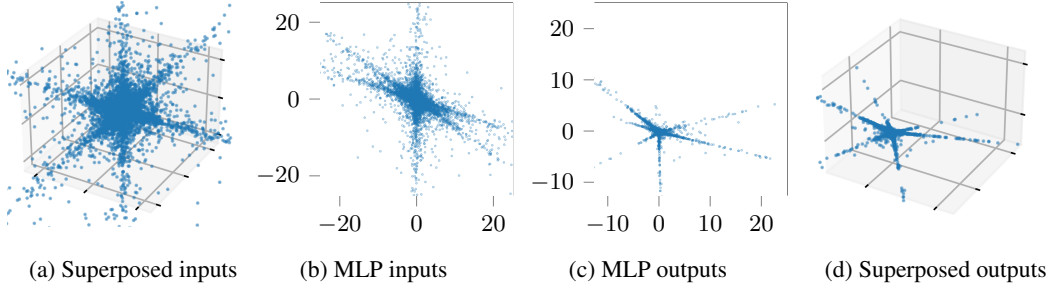


Figure 3: Input-versus-output comparison illustrating the effect of a randomly initialized neural network on superposed input data. We take 10K samples of $n_s = 3$ sparse input features from a Lomax distribution with shape $\alpha = 1$ and scale $\lambda = 1$ and project these to $n_d = 2$ dense input features by an i.i.d. standard normal matrix. We then pass the dense inputs through a two-layer MLP with ReLU activation and hidden size $4n_d$ and recover $n_s = 3$ sparse outputs via the inverse of the projection matrix. Comparing the superposed inputs (a) with the superposed outputs (d), the outputs are visibly sparser, indicating that the random network amplifies rather than merely preserves superposition.

4.1 MATRIX MULTIPLICATIONS PRESERVE SUPERPOSITION

We begin with the simplest case, a linear model, which makes the preservation of superposition exact rather than approximate: multiplication by a weight matrix W leaves the data superposed. Imagine that we generate superposed input data x by first generating n_s i.i.d. ‘sparse’ features z from a heavy-tailed Lomax distribution $z \sim \text{Lomax}(\alpha, \lambda)$. We can project the higher-dimensional, sparse z down to lower-dimensional, dense x with a matrix D , then add Gaussian noise with a small variance Σ , $x \sim \mathcal{N}(x; Dz, \Sigma)$. Importantly, if we multiply x by some matrix W , then $x' = Wx$ is *also* superposed: it is generated by the same model as x , except with different noise covariances and mappings from z to x , namely $x' \sim \mathcal{N}(z; WDz, W\Sigma W^T)$.

The same intuition extends to nonlinear networks, as a direct input-versus-output comparison makes visible when we pass the dense activations through a simple feed-forward network (MLP). Figure 3 shows an example with $n_s = 3$ sparse features projected down to $n_d = 2$ dense features: despite the nonlinearity, the MLP outputs remain visibly superposed. The comparison further points to amplification rather than mere preservation: relative to the inputs (Figures 3a and 3b), the outputs (Figures 3d and 3c) have noticeably fewer points between the ‘arms’, i.e., a sparser, more sharply superposed structure.

4.2 DO RANDOM NNS PRESERVE OR AMPLIFY SUPERPOSITION?

We tested this quantitatively by generating toy data following Sharkey et al. (2022), i.e., sampling ground-truth features on a hypersphere and generating correlated feature coefficients such that only a small number are active (Appendix I.1). We passed these inputs through a two-layer MLP at initialization and trained SAEs separately on the inputs and on the outputs, so the two can be compared on equal terms. The matched control was Gaussian-distributed data with the same mean and standard deviation as the superposed toy data, isolating superposition as the only difference. Throughout we used standard SAEs with an L^1 sparsity penalty (Appendix I.2).

Following Sharkey et al. (2022), we first confirmed near-perfect feature recovery on the inputs (Figure 4). We quantified recovery with the mean max cosine similarity (MMCS): for each data-generating feature we took its maximum cosine similarity to the SAE’s learned features (decoder weight vectors) and averaged over data-generating features. MMCS is only computable for the MLP inputs, where the ground-truth features are known, so for the outputs we used a different, comparable proxy: how easily an SAE reaches low reconstruction error at high sparsity, which increases with the degree of superposition. Concretely, we sweep the L^1 penalty coefficient to trace Pareto frontiers of explained variance against sparsity (Figure 5a), so a frontier lying up and to the left indicates more superposition.

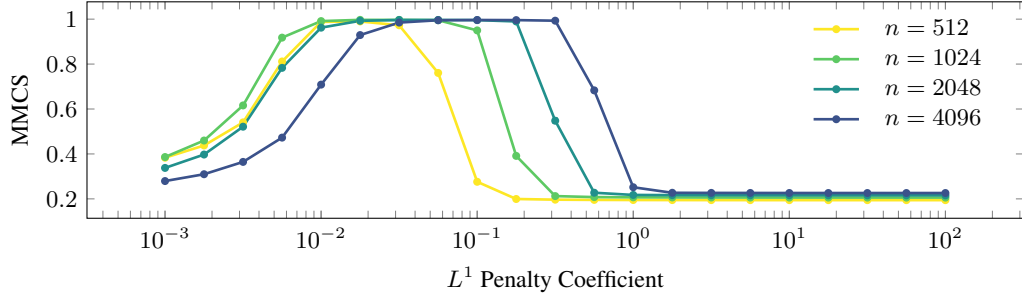
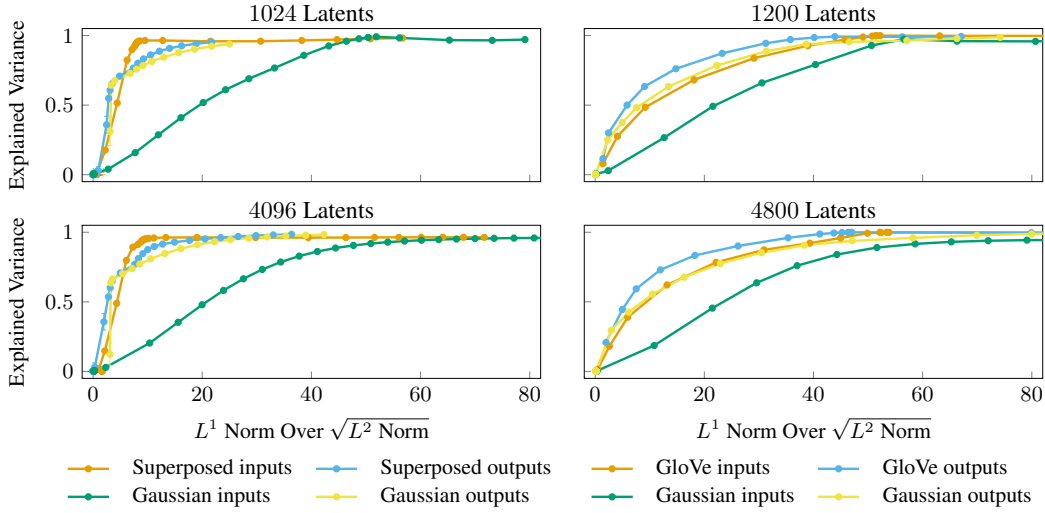


Figure 4: Mean max cosine similarity (MMCS) between the SAE’s learned features (decoder weight vectors) and the data-generating features, as a function of the L^1 penalty coefficient in the training loss, following Sharkey et al. (2022). Across an intermediate ‘Goldilocks zone’ of penalty coefficients, MMCS approaches its ceiling—i.e., the SAE recovers the data-generating features almost perfectly—provided there are enough latents to represent them.



(a) Toy datasets following Sharkey et al. (2022). (b) 300-dim. GloVe vectors (Pennington et al., 2014).

Figure 5: Pareto frontiers of explained variance against the L^1 norm divided by the square root of the L^2 norm (sparsity) for datasets that may exhibit superposition, matched Gaussian controls with the same mean and variance, and the corresponding outputs after passing each through a randomly initialized two-layer MLP (Section 4.2). Each point is one L^1 penalty coefficient; a frontier up and to the left means more sparsity at a given explained variance. The data-vs-control gap is wide at the input but nearly vanishes at the MLP output, showing that the random network sparsifies both distributions to a similar degree.

The input comparison behaves as expected and the margin is large: at any fixed level of explained variance, the SAE attains substantially greater sparsity for the superposed inputs than for the Gaussian control (Figure 5a; orange vs. blue-green). The output comparison is the revealing one: after passing both through the random MLP, the superposed and Gaussian outputs are separated by only a small margin, with the superposed outputs slightly sparser at a given explained variance. In other words, the large input-side gap nearly closes at the output, so the random MLP’s outputs are highly sparse almost regardless of the input distribution. We consider other sparsity measures and hyperparameters in Appendix I.

4.3 DO TOKEN EMBEDDINGS EXHIBIT SUPERPOSITION?

If randomly initialized networks preserve or amplify superposition, then the near-parity of Section 3 would follow to the extent that the inputs to transformer language models are already superposed. We test this directly by rerunning the Section 4.2 comparison on language data. Concretely, we train SAEs on four matched groups—pre-trained GloVe word vectors, the embedding matrices of Pythia models, the outputs of passing each through a randomly initialized two-layer MLP, and Gaussian controls—and compare their Pareto frontiers. The setup is identical to Section 4.2, except that the number of data points is fixed by the number of word embeddings or tokens and we use a single random seed.

Comparing frontiers, the input-side gap between the GloVe word vectors and their Gaussian control (Figure 5b) is smaller than the corresponding gap for the toy superposed datasets of Section 4.2 (Figure 5a)—consistent with real embeddings being less starkly superposed than the toy data. More tellingly, the same amplification pattern recurs: for both the GloVe vectors and their control, the Pareto frontier shifts up and to the left after passing through the random MLP, reinforcing that random networks ‘sparsify’ their inputs (i.e., increase apparent superposition) largely irrespective of the input distribution.

5 LIMITATIONS

Our headline finding is quantitative—across the Pythia suite, aggregate auto-interpretability measures produce comparable scores for SAEs on trained and on randomly initialized models, even though the random models are unlikely to host computationally interesting features. Because testing every dataset and architecture is infeasible, we deliberately anchored the comparison to a widely used, well-controlled setting: the Pythia family, heavily adopted in mechanistic interpretability research (e.g., Paulo and Belrose, 2025; Ghilardi et al., 2025; Mueller, 2024), and the RedPajama-V2 dataset, representative of typical pre-training data. This spans two orders of magnitude in model size but a single model family, so the exact magnitude of the trained-vs-random gap may differ elsewhere. We also held the explainer fixed at the default model of the EleutherAI auto-interpretability framework (Caden et al., 2025); a stronger explainer could raise absolute scores, though we would expect it to lift the trained and randomized variants together and thus leave the small gap between them largely unchanged. Finally, we stress the precise scope of the quantitative claim: we do not assert that SAEs extract no more information from trained than from randomized transformers, only that a high aggregate auto-interpretability score does not, by itself, evidence interesting underlying features.

6 CONCLUSION

We ran a controlled, like-for-like comparison of SAEs on trained and on randomly initialized transformers across the Pythia suite, evaluated with a common suite of quantitative metrics. The central empirical result is a measured near-parity: on aggregate metrics—above all auto-interpretability scores—the trained and randomized variants score close together and far from the chance-level control, so the trained-vs-random margin these metrics ought to expose is small or absent. The qualitative distinction we expected is real but the aggregate numbers do not register it: features from trained transformers grow more complex and abstract with depth, yet the aggregate scores stay flat across variants.

This does not imply that SAEs on real models fail to learn meaningful computational features; it pinpoints a discriminative-power gap in current evaluation. A high aggregate auto-interpretability score is not sufficient evidence for the discovery of complex, learned computations, because the identical score is reachable from random weights that only preserve or amplify structure already present in the data or architecture. Our token-distribution entropy analysis, while preliminary, is a concrete proof-of-concept that the gap is closable: it recovers a trained-vs-random difference in feature ‘abstractness’ along the depth axis that the aggregate scores miss entirely. Future work should build metrics with demonstrable power to separate learned computation from such baselines and to quantify the computational significance of discovered features. Above all, our results make the case for reporting randomized null models alongside trained ones as a matter of routine: only against such

540 a baseline can a metric's trained-vs-random margin be read off directly, and only then can discovered
541 features be confidently attributed to learning.
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A BROADER IMPACT

This work investigates a method currently used for mechanistic interpretability of LLMs, yielding results that challenge certain assumptions about sparse autoencoders. By showing quantitatively that SAEs reach comparable aggregate auto-interpretability scores on both random and trained transformers—with the two settings separated by only a small margin—our findings raise important questions about what these SAE evaluation methods are actually capturing.

By better understanding the metrics of SAE quality, we hope that this work will contribute to a more informed search of better SAE-like methods and thus help to make these models more interpretable and to mitigate the potential harm these models could cause. Since our work is an empirical study of the capabilities of a presently used method, and it shows that the method provides interpretation of both random and trained transformers, we think the risk that this work could lead to negative social impact is minimal.

B AUTO-INTERPRETABILITY ROC CURVES

Figures 6, 8, and 12 extend the trained-vs-random ‘fuzzing’ AUROC comparison to the 70M, 160M, and 1B models, and Figures 7, 9, and 13 do the same for ‘detection’ AUROC. Across both scoring methods and all three sizes, the trained and randomized curves stay close together, confirming that the near-parity of the main text is not specific to one scoring rule or one model size.

B.1 PYTHIA 70M

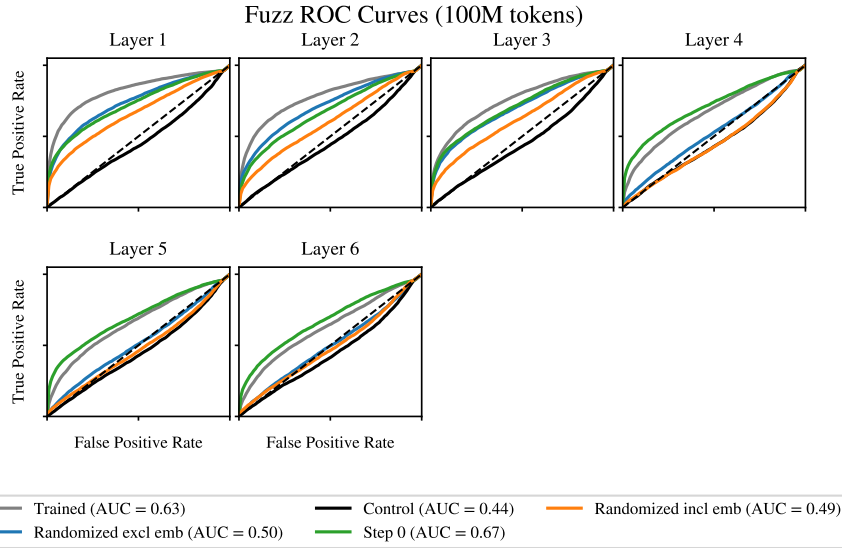


Figure 6: Trained-vs-random ‘fuzzing’ AUROC for Pythia-70m (100 latents per SAE). Even at the smallest scale the trained and randomized ROC curves lie close together at each layer, and both curves shift downward together as the layer index increases.

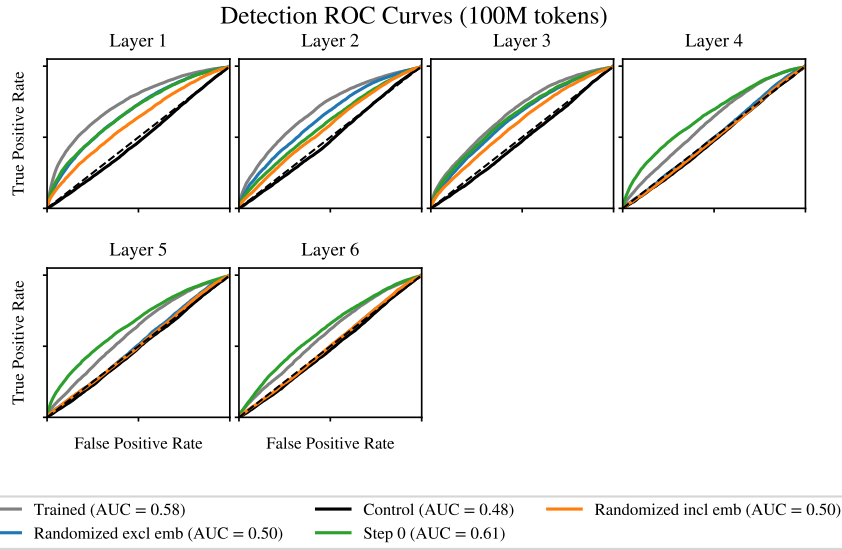


Figure 7: Trained-vs-random ‘detection’ AUROC for Pythia-70m (100 latents per SAE). As with fuzzing, the variants track one another closely at each layer, and performance declines together with increasing layer index.

B.2 PYTHIA 160M

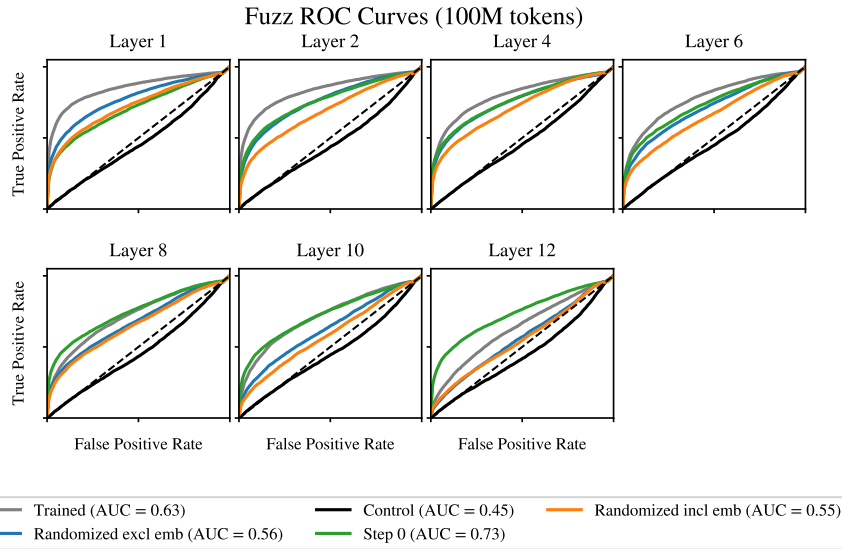


Figure 8: Trained-vs-random ‘fuzzing’ AUROC for Pythia-160m (100 latents per SAE). The trained and randomized curves remain closely matched layer by layer, with a shared downward trend as the layer index increases.

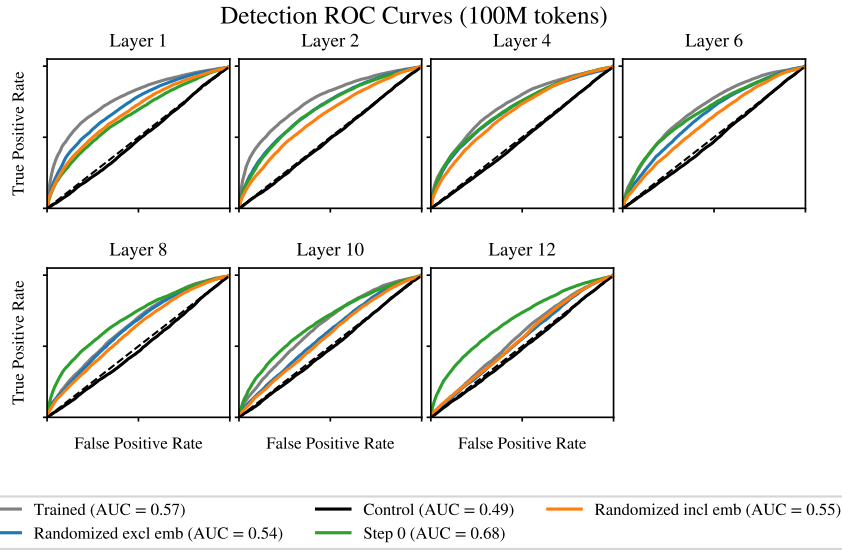


Figure 9: Trained-vs-random ‘detection’ AUROC for Pythia-160m (100 latents per SAE). The variants stay close together at each layer, and performance degrades together with increasing layer index.

B.3 PYTHIA 410M

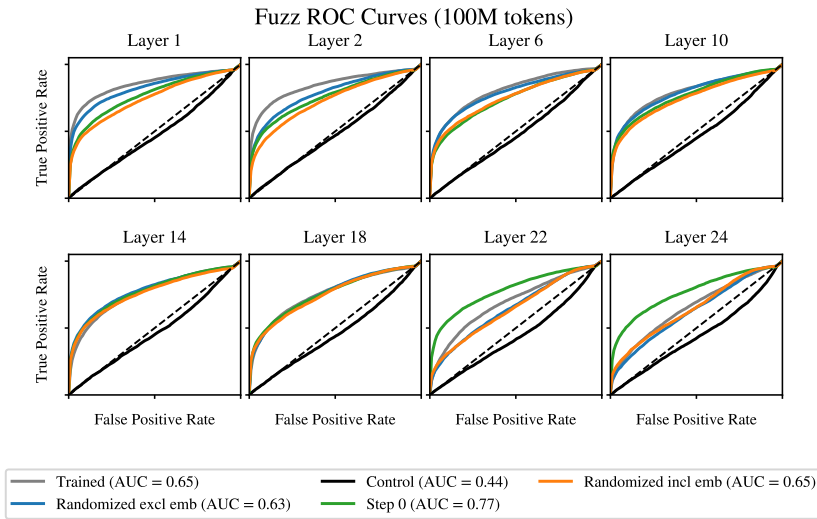


Figure 10: Trained-vs-random ‘fuzzing’ AUROC for Pythia-410m (100 latents per SAE). The trained and randomized curves are essentially indistinguishable at every layer—so at this scale fuzzing AUROC does not separate trained from randomized models—and both decline together with increasing layer index.

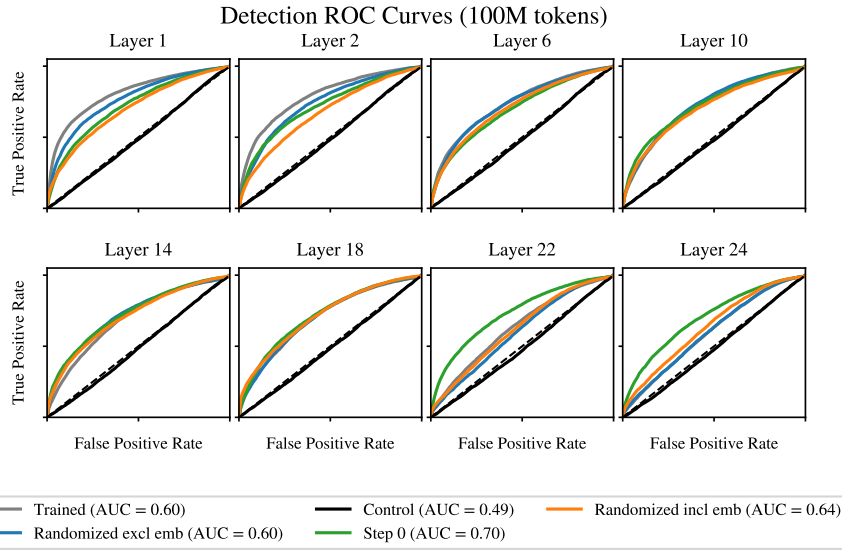


Figure 11: Trained-vs-random ‘detection’ AUROC for Pythia-410m (100 latents per SAE). The variants remain closely matched at each layer, with a shared decline as the layer index increases.

B.4 PYTHIA-1B

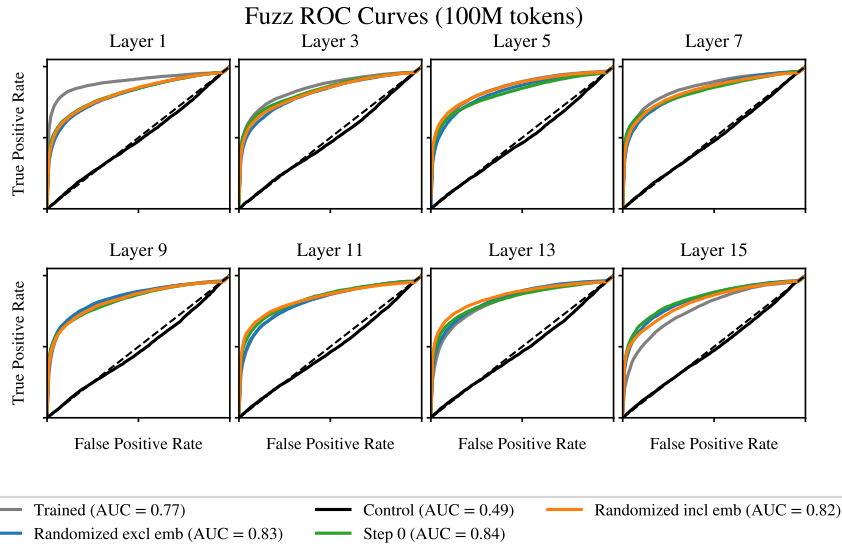


Figure 12: Trained-vs-random ‘fuzzing’ AUROC for Pythia-1b (100 latents per SAE). The trained and randomized curves are closely matched at every layer; unlike the smaller models, here the curves do not decline with layer index.

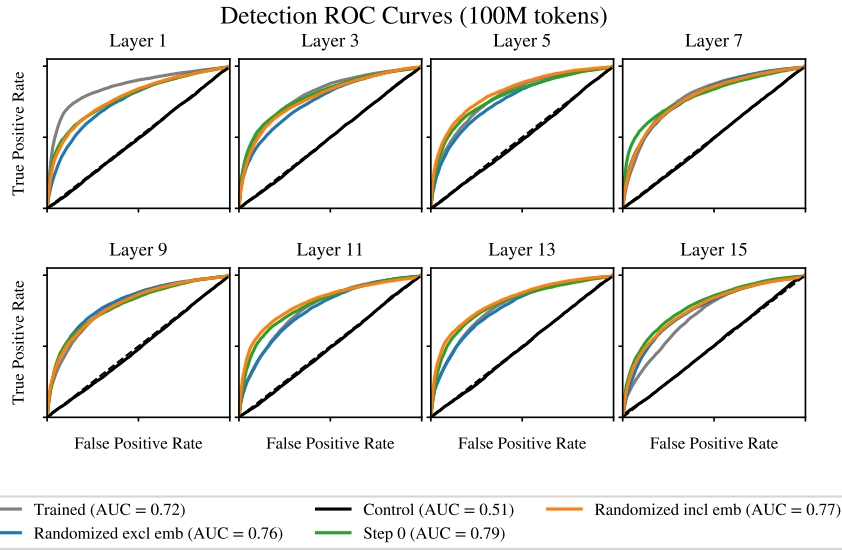


Figure 13: Trained-vs-random ‘detection’ AUROC for Pythia-1b (100 latents per SAE). The variants track one another closely at each layer; as with fuzzing at this scale, the curves show no overall decline with layer index.

B.5 PYTHIA 6.9B

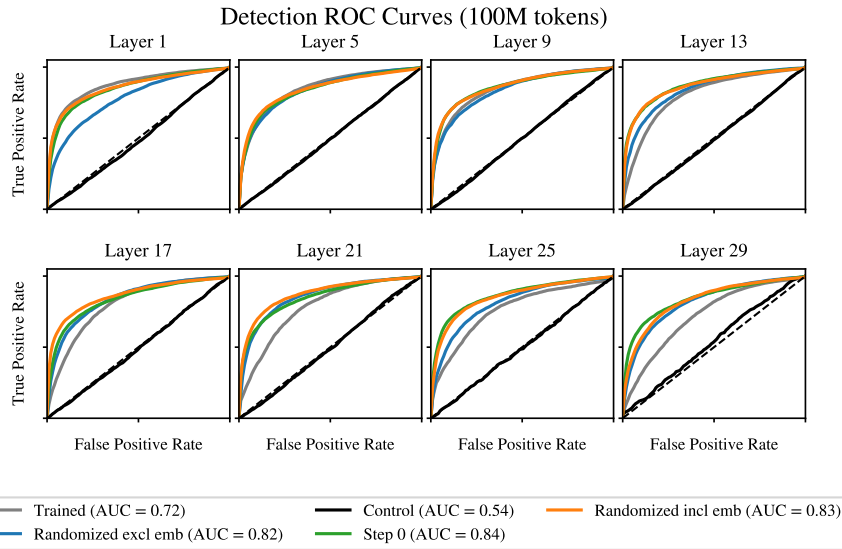


Figure 14: Trained-vs-random ‘detection’ AUROC for Pythia-6.9b (100 latents per SAE). As with fuzzing at this scale (Figure 1), the trained and randomized curves are nearly coincident at every layer, confirming the near-parity under the alternative scoring rule.

C EFFECT OF INCREASED TRAINING DATA

Our primary experiments used SAEs trained on 100M tokens (Section 3). To rule out insufficient training data as the cause of the observed near-parity, we retrained a subset of SAEs on $10\times$ the data—1B tokens from RedPajama—and re-ran the same comparison (Figure 15). The trained-vs-random gap remains just as small, so the effect is not an artifact of under-training.

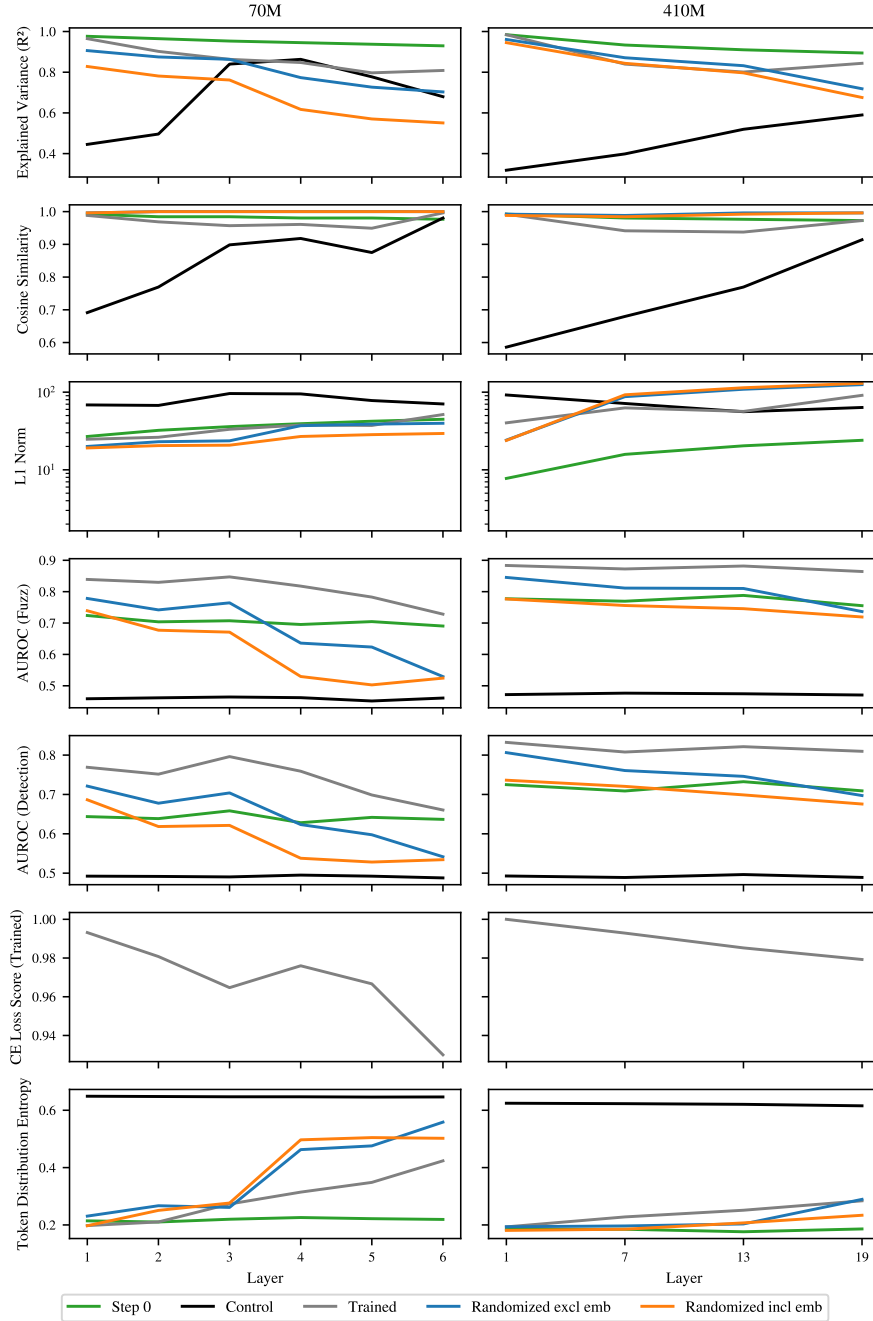


Figure 15: Evaluation metrics for SAEs trained with one billion tokens on Pythia-70m and 410m. Compared column-for-column against the 100M-token results in Figure 2, the curves are qualitatively unchanged: the trained-vs-random near-parity holds at $10\times$ the training data.

D EFFECT OF DECREASED TRAINING DATA FOR PYTHIA-1B

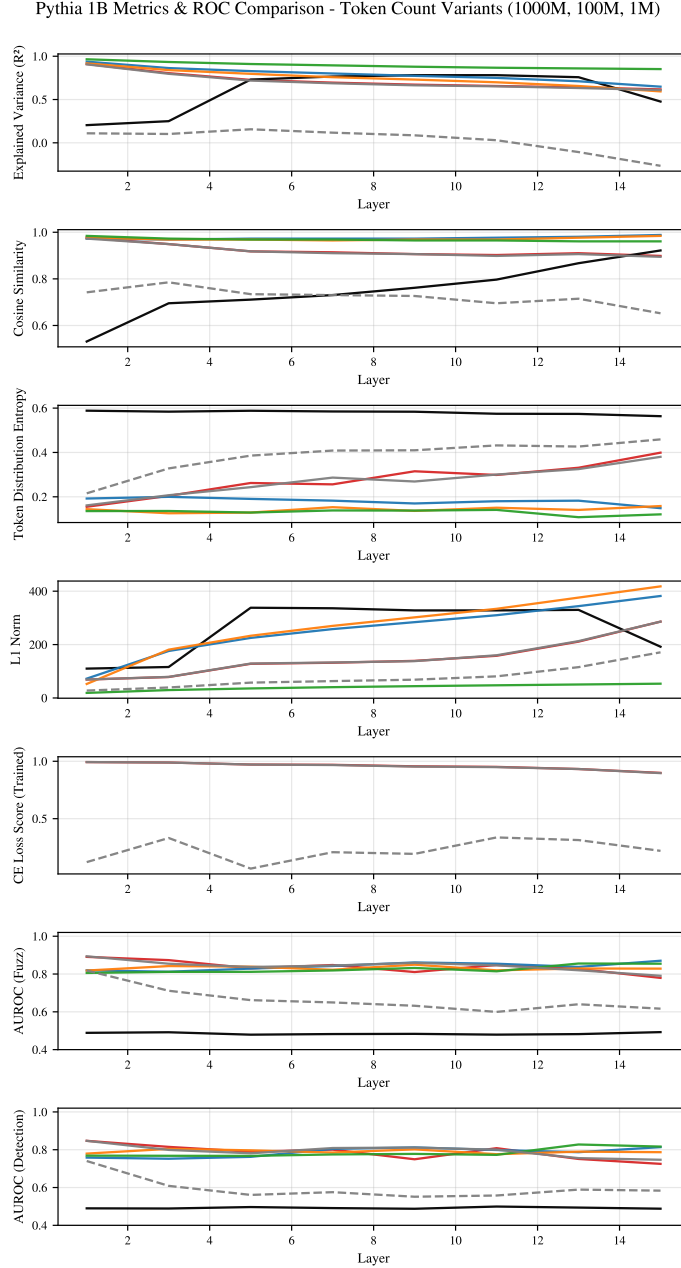
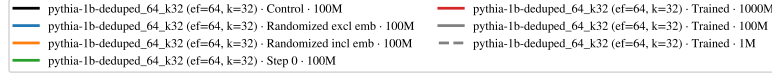


Figure 16: Evaluation metrics for SAEs trained with 1M versus 1B tokens on Pythia-1b. The $1000\times$ reduction in data has a large, expected effect on reconstruction—explained variance and CE loss score are markedly lower at 1M tokens, the signature of under-training—and a smaller effect on auto-interpretability, which is only slightly lower at the earliest layers but falls off sharply with depth. Even so, the layer-wise trends in auto-interpretability and token-distribution entropy match those of the other SAEs, so the qualitative comparison is preserved.

1188 E UNCERTAINTY PLOTS FOR PYTHIA-70M
1189

1190 To confirm that the near-parity is not an artifact of seed noise, we recomputed every evaluation metric
1191 on Pythia-70m over five random seeds and report the resulting uncertainty.
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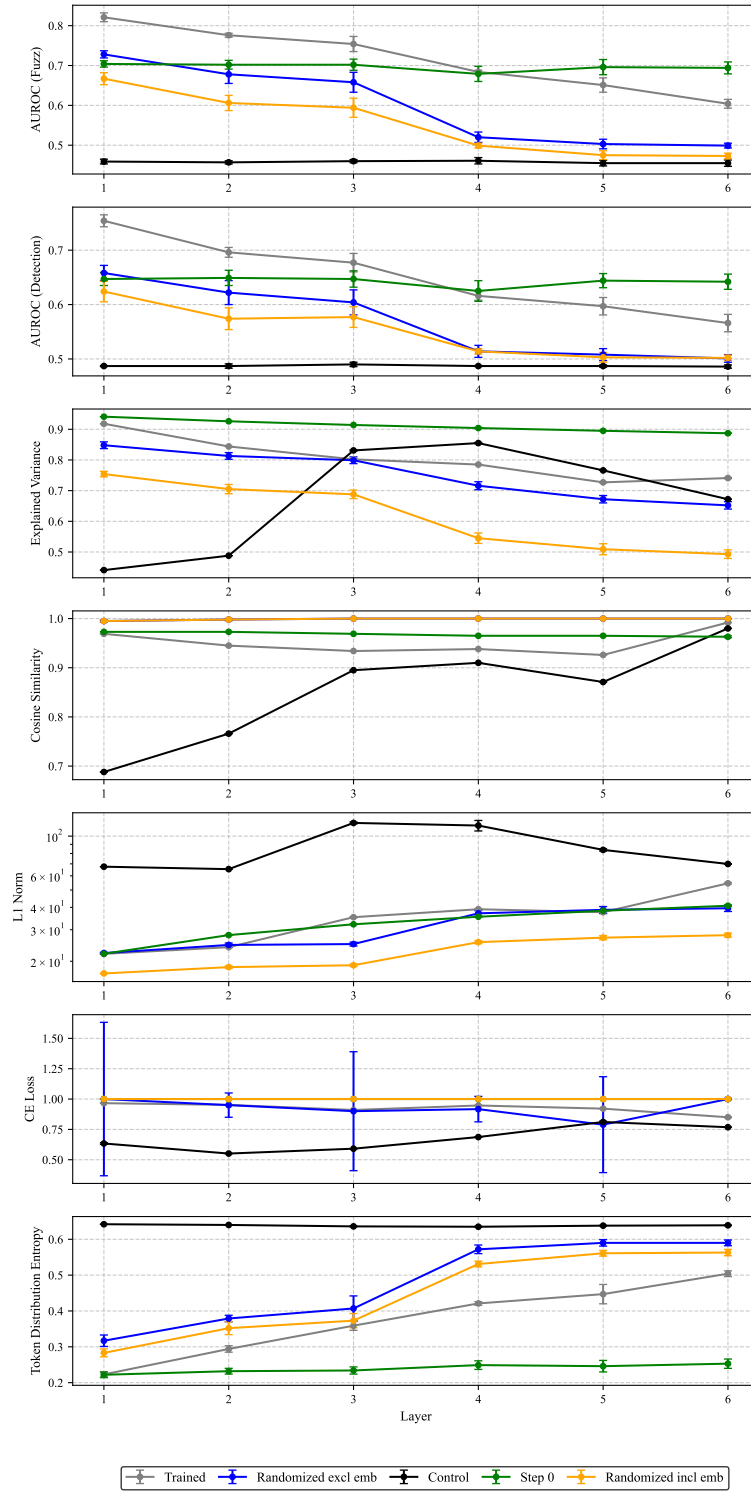


Figure 17: Pythia-70m metrics with uncertainty over five random seeds. The between-variant gaps—small between trained and randomized, large against the control—exceed the seed-to-seed variability, so the near-parity reported in the main text is not attributable to seed noise.

F EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-160M

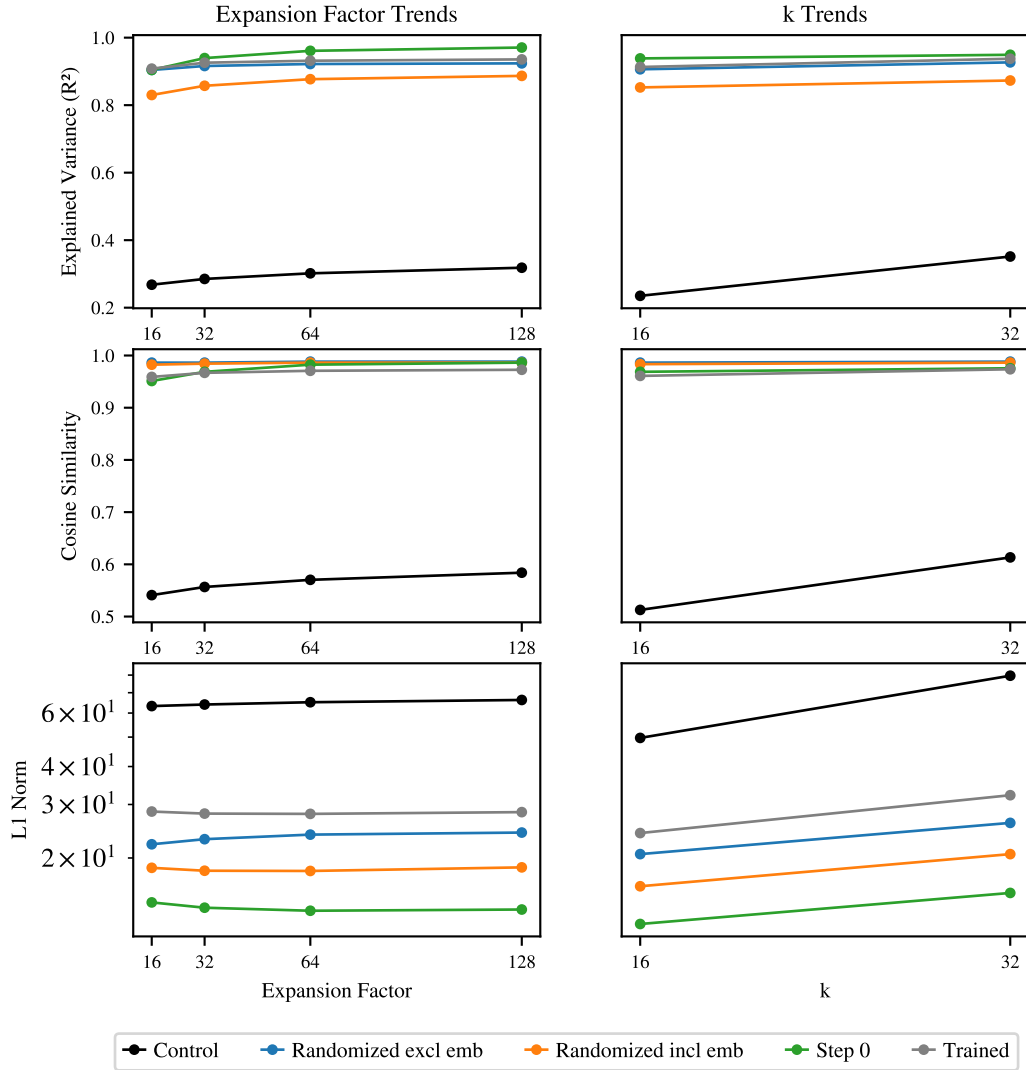


Figure 18: SAE performance is stable across the hyperparameter sweep, and the variant ordering is preserved throughout. Over expansion factors R from 16 to 128 and sparsities k of 16 and 32, the standard evaluation metrics move little and every initialization strategy keeps its relative ranking (trained and randomized close together, control apart). The comparison therefore does not hinge on hyperparameters, and our moderate defaults ($R = 64$, $k = 32$) are representative.

G EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-1B

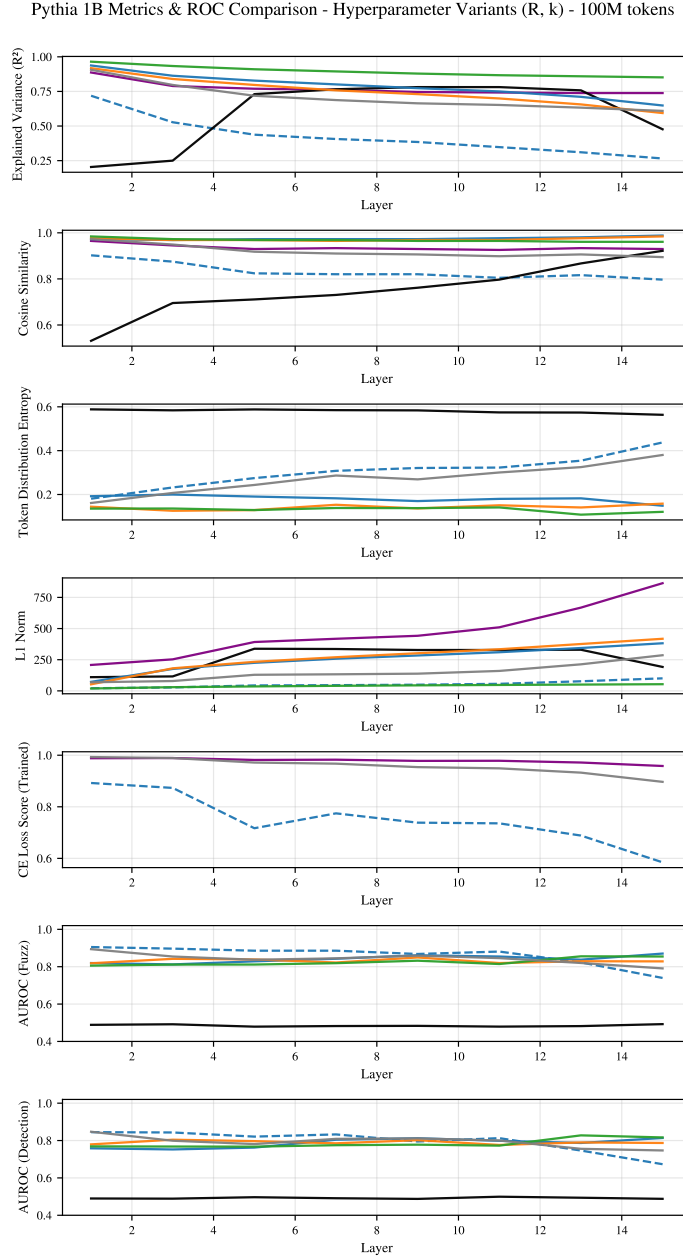


Figure 19: Evaluation metrics for SAEs on Pythia-1b under different hyperparameters, alongside the default-hyperparameter results from Figure 2. A deliberately undersized SAE ($R = 2, k = 4$) is sharply separated from the defaults on explained variance and CE loss score—so those reconstruction metrics do respond strongly to capacity. Yet its auto-interpretability scores remain close to those of the default SAEs on both trained and randomized models, underscoring that auto-interpretability is comparatively insensitive to the underlying model and to capacity.

H TOKEN DISTRIBUTION ENTROPY VS. AUTO-INTERPRETABILITY

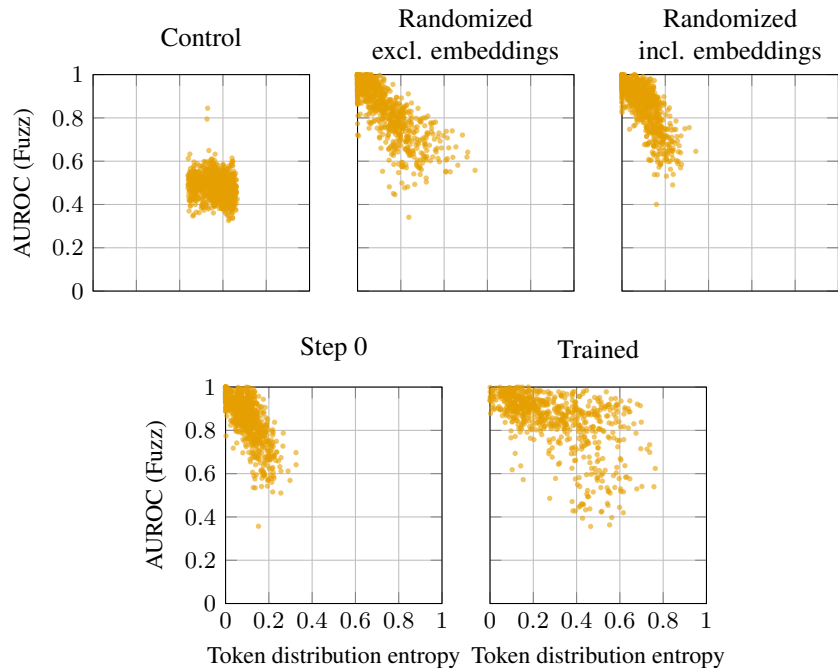


Figure 20: Per-latent token-distribution entropy versus ‘fuzzing’ AUROC for SAEs across multiple layers of Pythia-6.9b. Each point is a single latent, drawn from the same sample used for the aggregate metrics in Figures 1 and 2. Whereas the aggregate scores gave near-identical numbers across variants, the joint entropy–AUROC distribution pulls the variants apart: only the trained variant populates the high-entropy, high-AUROC region.

Viewed jointly rather than as aggregates, entropy and AUROC cleanly separate the three regimes (Figure 20):

- **Control:** Latents cluster at high entropy (activation patterns spread across many tokens) and low AUROC (explanations that fail to account for those patterns), with no discernible correlation between the two.
- **Randomized:** Under every randomization scheme of Section 3, entropy and AUROC are negatively correlated: the more tokens a latent fires on, the worse its explanation predicts its activations. The randomized points stay confined to the low-entropy side.
- **Trained:** The correlation is weaker, and—decisively—the trained variant is the only one to populate the high-entropy, high-AUROC corner. These are latents that fire across many tokens yet remain well explained, i.e., genuinely multi-token semantic features absent from every randomized variant.

This is the sharpest single view of what the aggregate scores hide: averaged over latents, trained and randomized SAEs look alike, but the joint distribution reveals a class of high-entropy, well-explained features unique to the trained model. Randomized models with consistent token embeddings readily produce easy-to-explain single-token features, but only transformers trained on natural language produce the more complex semantic features that occupy the upper-right region.

I A TOY MODEL OF SUPERPOSITION

Section 4 reported comparisons on toy data designed to exhibit superposition (Sharkey et al., 2022) and on GloVe word vectors (Pennington et al., 2014). Here we give the data-generation procedure and training setup in full, so that the reported comparisons can be reproduced exactly.

I.1 DATA GENERATION

First, we construct ground-truth features by sampling n_s points on an n_d -dimensional hypersphere.

For each sample, we determine the feature coefficients by generating $A \in \mathbb{R}^{n_s \times n_s}$ where $A_{ij} \sim \mathcal{N}(0, 1)$, defining a covariance matrix $\Sigma = AA^T$, sampling $\vec{\alpha} \in \mathbb{R}^{n_s}$ where $\alpha_i \sim \mathcal{N}(\vec{0}, \Sigma)$, projecting α_i onto the c.d.f. of $\mathcal{N}(0, 1)$, decaying $\alpha_i \rightarrow \alpha_i^{\lambda i}$ where $\lambda \in \mathbb{R}$, normalizing $\alpha_i \rightarrow m\alpha_i / n_s \sum_j \alpha_j$ where $m \in \mathbb{R}$, and performing n_s independent Bernoulli trials with $p = \alpha_i$. Finally, we multiply the trial outcomes by n_s independent samples from a continuous uniform distribution $\mathcal{U}_{[0,1]}$.

The parameter λ determines how sharply the frequency of nonzero ground-truth feature coefficients decays with the feature index i . The parameter m is the expected value of the number of nonzero feature coefficients for each sample.

To match Sharkey et al. (2022) exactly, we set $n_s = 512$, $n_d = 256$, $\lambda = 0.99$, and $m = 5$. A Python implementation of the full procedure is given in Figure 21.

I.2 TRAINING

The SAEs described in Section 4 comprise a linear encoder with a bias term, a ReLU activation function, and a linear decoder without a bias term. We use orthogonal initialization for the decoder weights and normalize the decoder weight vectors before each training step.

The training loss is the mean squared error (MSE) between the input and decoded vectors plus the mean L^1 norm of the encoded vectors scaled by a coefficient; sweeping this coefficient over more than five orders of magnitude, from 1×10^{-3} to 100, is what traces each Pareto frontier.

For the toy data, we train for 100 epochs on 10K data points and average over 10 random seeds; for the word vectors, we train for 100 epochs on 400K data points with a single seed. In both cases, all reported evaluation metrics are computed on a held-out 10% validation split.

The MLPs described in Section 4 comprise two layers (i.e., one hidden layer) and a ReLU activation function. The input and output sizes are both equal to n_d , and the hidden size is $4n_d$. We loosely based these choices on the feed-forward network components of transformer language models.

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1524 1 def generate_sharkey(
1525 2     num_samples: int,
1526 3     num_inputs: int,
1527 4     num_features: int,
1528 5     avg_active_features: float,
1529 6     lambda_decay: float,
1530 7 ) -> tuple[Tensor, Tensor]:
1531 8     """
1532 9     Args:
1533 10         num_samples (int): The number of samples to generate.
1534 11         num_inputs (int): The number of input dimensions.
1535 12         num_features (int): The number of ground truth features.
1536 13         avg_active_features (float): The average number of
1537 14             ground truth features active at a time.
1538 15         lambda_decay (float): The exponential decay factor for
1539 16             feature probabilities.
1540 17     """
1541 18     features = torch.randn(num_inputs, num_features)
1542 19     features /= torch.norm(features, dim=0, keepdim=True)
1543 20
1544 21     covariance = torch.randn(num_features, num_features)
1545 22     covariance = covariance @ covariance.T
1546 23     correlated_normal = MultivariateNormal(
1547 24         torch.zeros(num_features), covariance_matrix=covariance
1548 25     )
1549 26
1550 27     samples = []
1551 28     for _ in range(num_samples):
1552 29         p = STANDARD_NORMAL.cdf(correlated_normal.sample())
1553 30         p = p ** (lambda_decay * torch.arange(num_features))
1554 31         p = p * (avg_active_features / (num_features * p.mean()))
1555 32         p = torch.bernoulli(p.clamp(0, 1))
1556 33         coef = p * torch.rand(num_features)
1557 34
1558 35         sample = coef @ features.T
1559 36         samples.append(sample)
1560 37
1561 38     return torch.stack(samples), features

```

Figure 21: Python implementation of the data-generation procedure of Sharkey et al. (2022), used to produce the superposed toy datasets compared in Section 4.

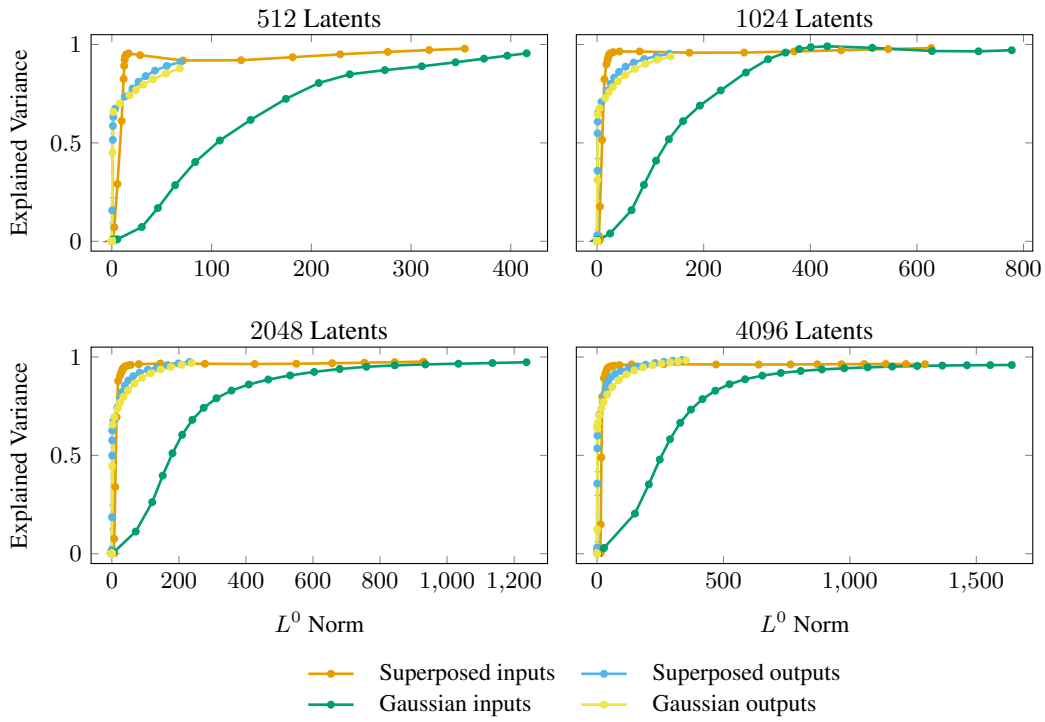


Figure 22: Explained-variance-vs- L^0 -norm (sparsity) Pareto frontiers comparing toy datasets generated to exhibit superposition, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. The wide superposed-vs-control gap at the input narrows sharply at the output, the same amplification pattern seen under the $L^1/\sqrt{L^2}$ sparsity measure.

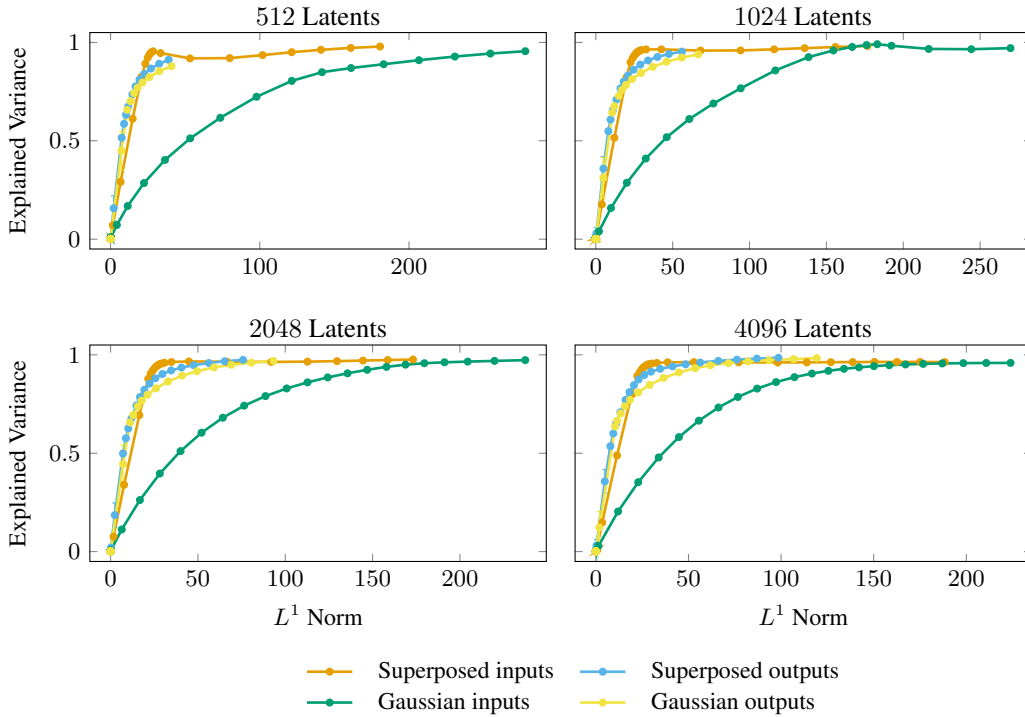


Figure 23: Explained-variance-vs- L^1 -norm (sparsity) Pareto frontiers comparing toy datasets generated to exhibit superposition, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. As under the other sparsity measures, the large input-side superposed-vs-control gap nearly closes at the MLP output.

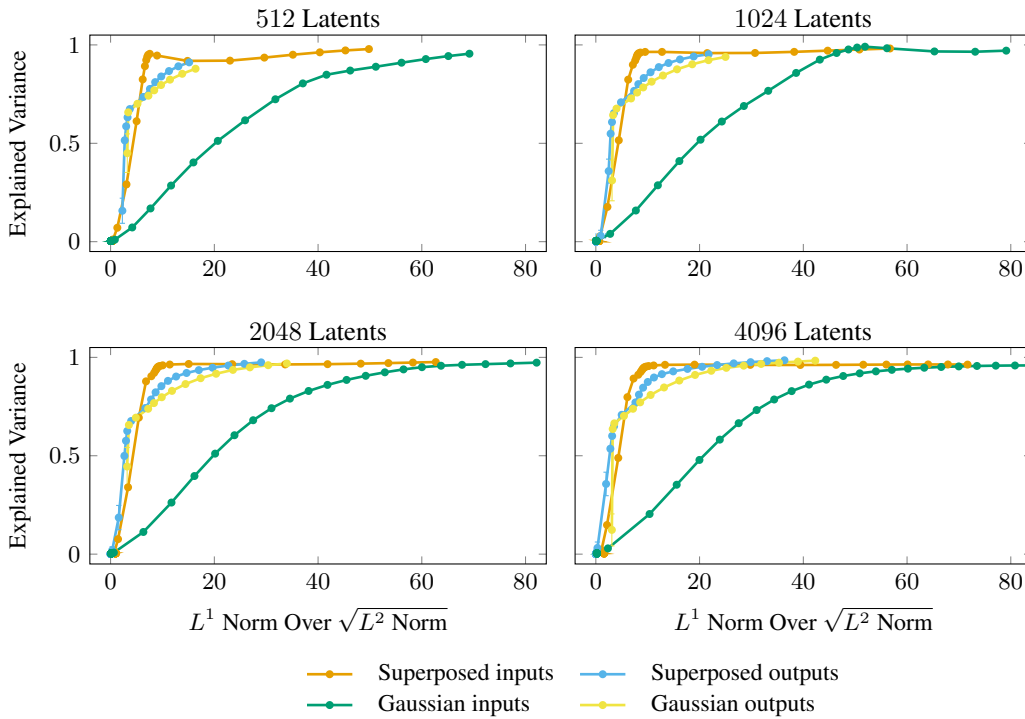


Figure 24: Explained-variance-vs- $L^1/\sqrt{L^2}$ (sparsity) Pareto frontiers comparing toy datasets generated to exhibit superposition, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. This is the full-page version of the comparison summarized in Figure 5a, again showing a wide input-side gap that all but disappears at the output.

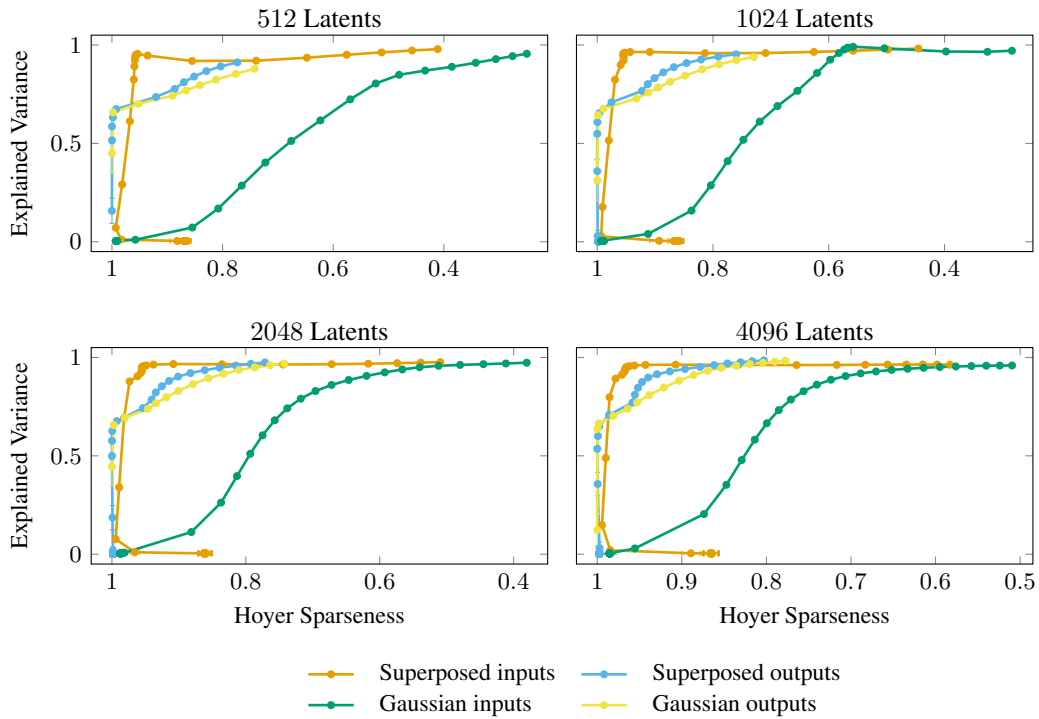


Figure 25: Explained-variance-vs-Hoyer-sparseness Pareto frontiers comparing toy datasets generated to exhibit superposition, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. The same input-to-output narrowing of the superposed-vs-control gap holds under this fourth sparsity measure.

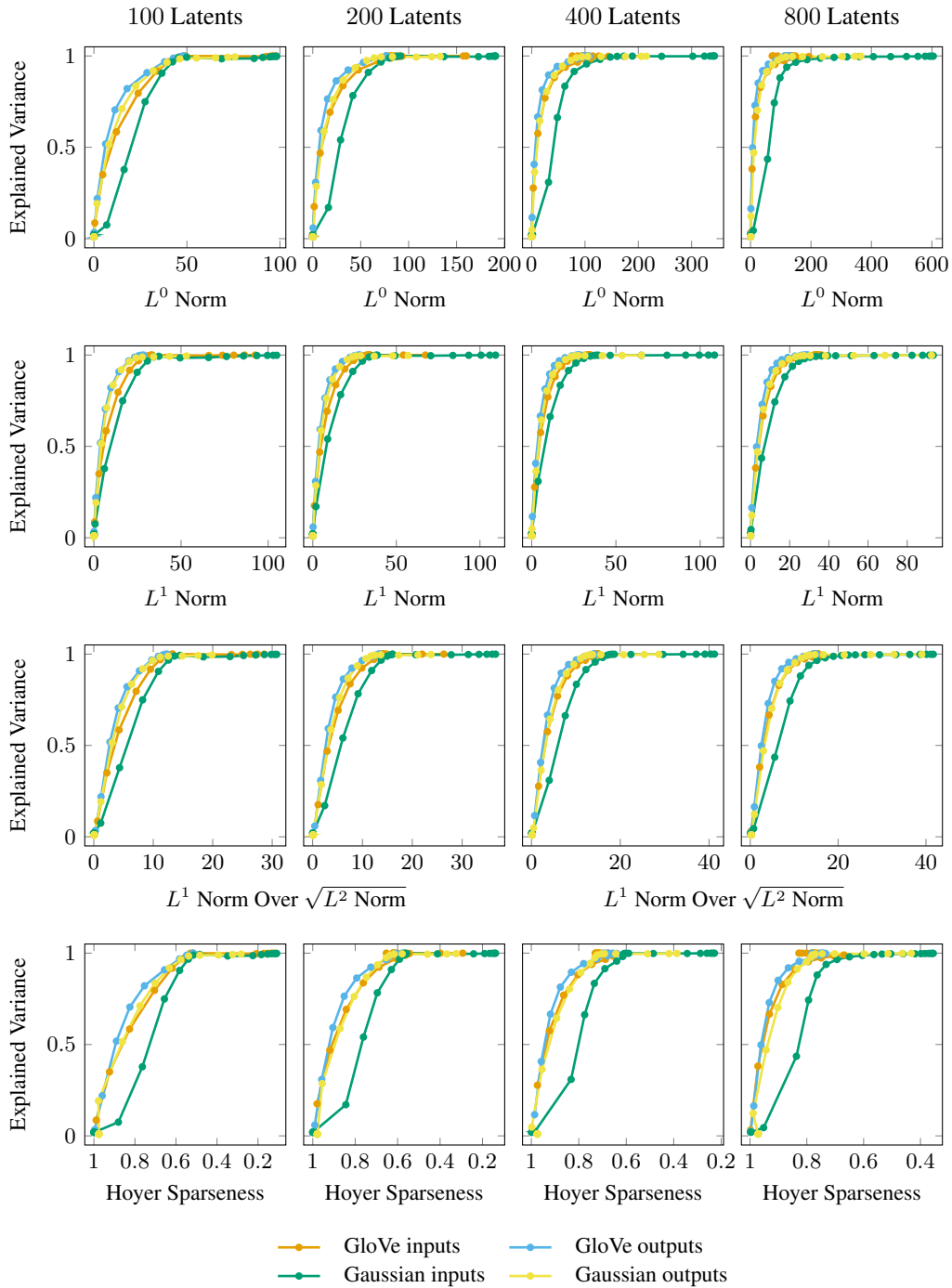


Figure 26: Explained-variance-vs-sparsity Pareto frontiers comparing 50-dimensional GloVe word vectors, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. As at higher dimensions, the modest input-side GloVe-vs-control gap shrinks further at the output.

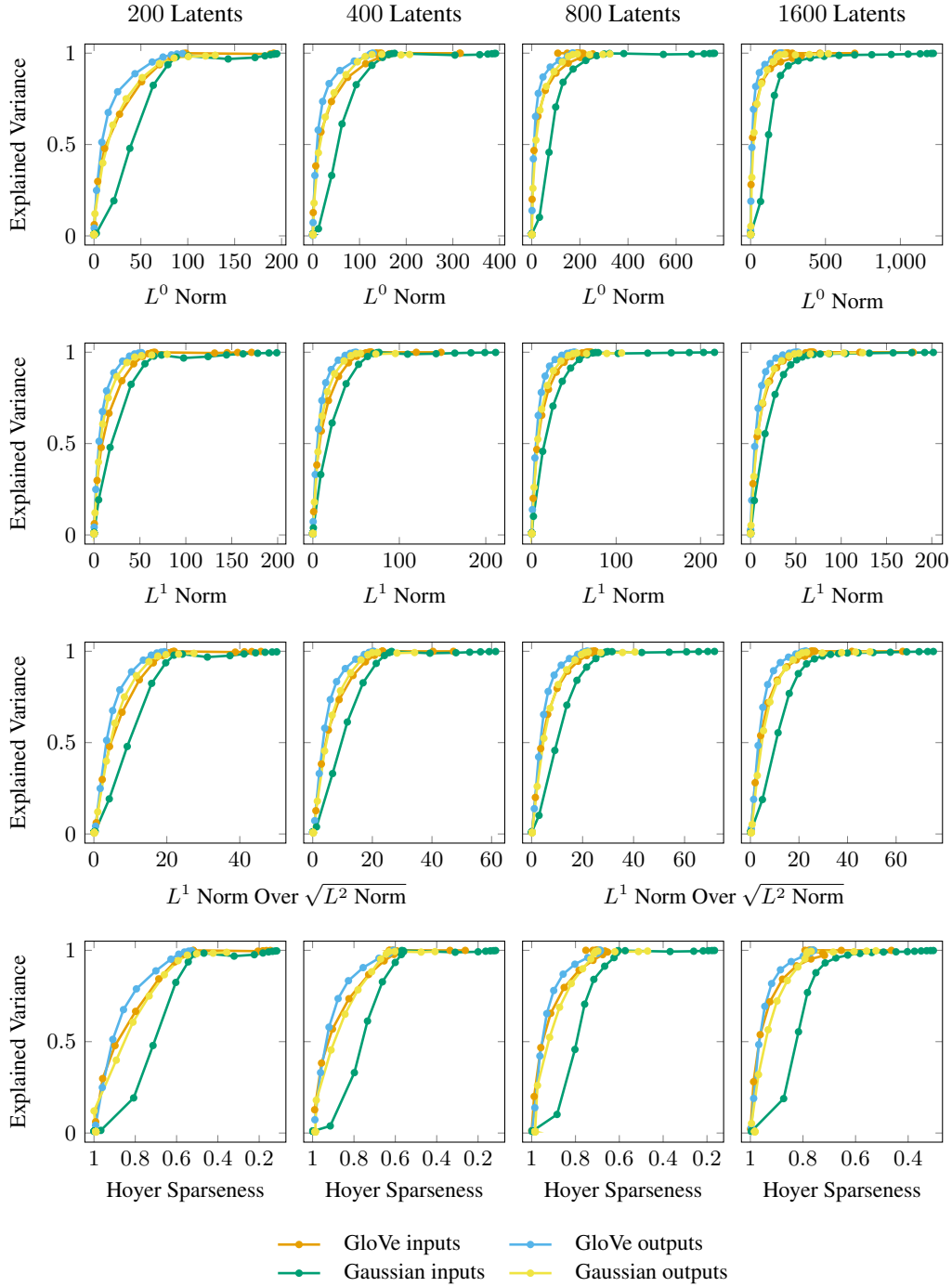


Figure 27: Explained-variance-vs-sparsity Pareto frontiers comparing 100-dimensional GloVe word embeddings, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. The GloVe-vs-control gap is small at the input and narrower still at the output.

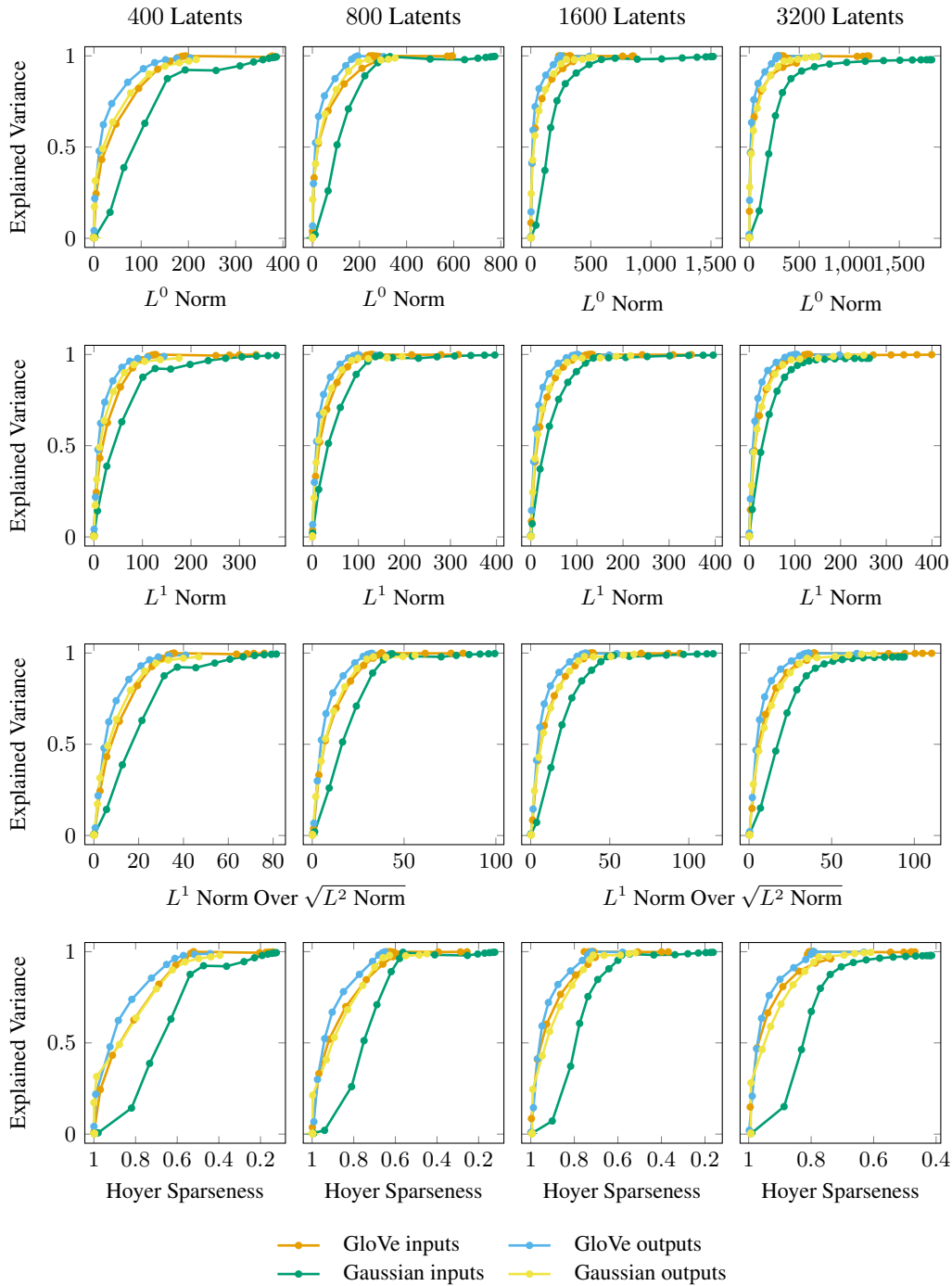


Figure 28: Explained-variance-vs-sparsity Pareto frontiers comparing 200-dimensional GloVe word embeddings, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. The comparison mirrors the other dimensions: a small input-side GloVe-vs-control gap that narrows at the output.

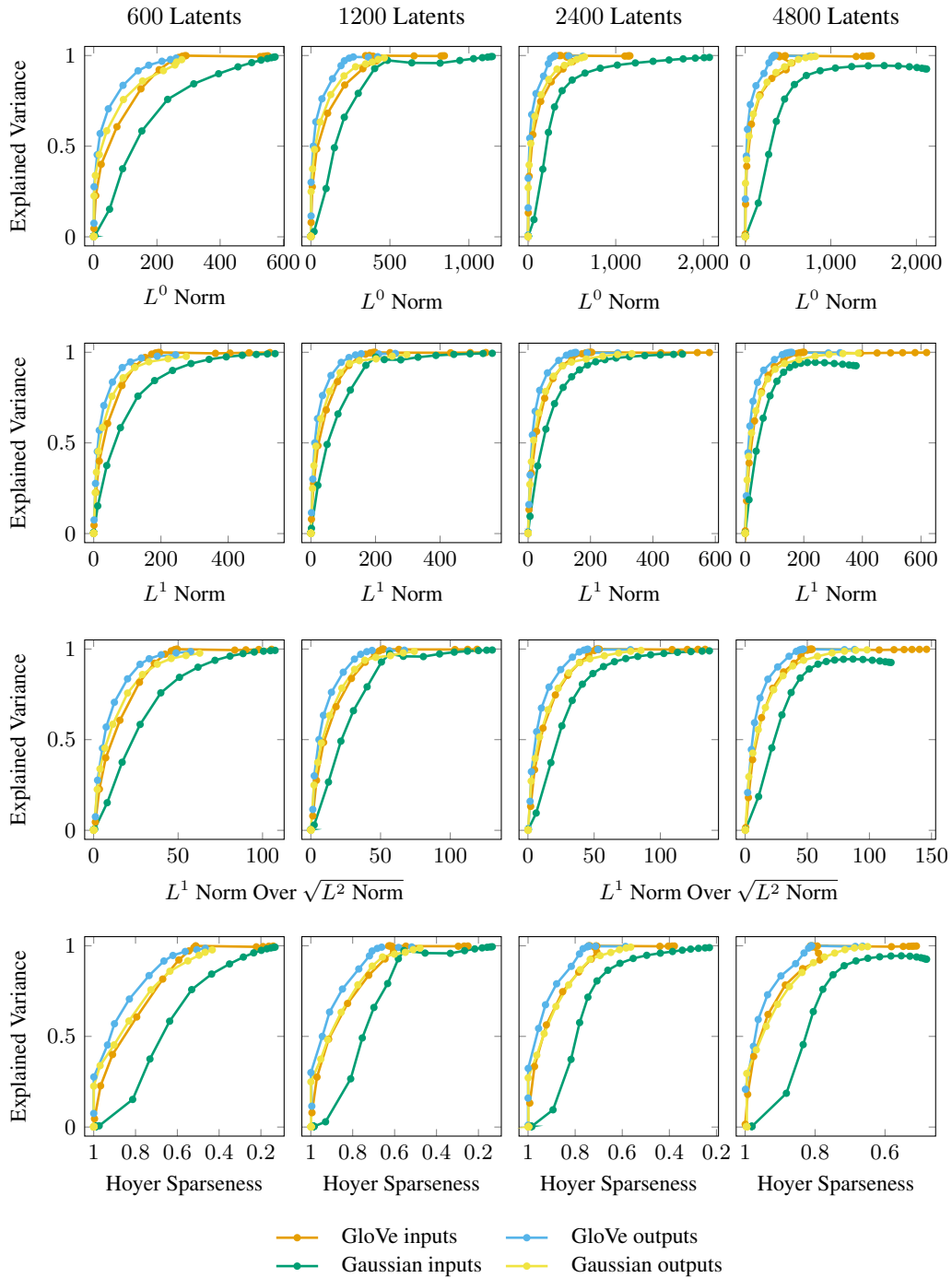


Figure 29: Explained-variance-vs-sparsity Pareto frontiers comparing 300-dimensional GloVe word embeddings, matched Gaussian controls with the same mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP. Consistent with the lower dimensions and with Figure 5b, the GloVe-vs-control gap is small at the input and narrows at the output.

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J EXAMPLE FEATURES FOR PYTHIA-6.9B

VARIANT: TRAINED

FEATURE 180935 (LAYER 0)

Interpretation: The term "security" is predominantly used to refer to protection, safety, and measures to prevent harm, while "oz" is likely referring to ounces, possibly in a context of measurement or quantification, although "oz" appears less frequently and often in a different context.

Top Examples:

1. Text: <endoftext—ζ—. If for any reason you are unhappy with our service please contact us directly so we can make it right for you. Journal of Cyber Security, Vol.
Activation: 4.3750
Active tokens: Security
2. Text: <endoftext—ζ— Security Practitioner. Tremendously passing CompTIA Advanced Security Practitioner (casp) cert has never been as easy as it
Activation: 4.3125
Active tokens: Security Security
3. Text: in trusted hands for your Cyber Security career or staffing needs. Call 0203 643 0248 to find out more. Technically proficient using
Activation: 4.3125
Active tokens: Security

FEATURE 93790 (LAYER 8)

Interpretation: Nouns and phrases related to economic concepts, development, and business, often referring to growth, progress, and improvement.

Top Examples:

1. Text: training requirements. See "Workforce" section for additional information. The Economic Development Transportation Fund, commonly referred to as the "Road Fund," is an
Activation: 21.3750
Active tokens: Development
2. Text: Montréal. The Williamsburg Economic Development Authority offers a 33% matching grant up to \$7,500 for exterior improvements to existing businesses in the City of
Activation: 20.2500
Active tokens: Development Authority
3. Text: Correction: In a July 16 web story The Real Deal incorrectly stated that the Economic Development Corporation was "circumventing" laws with its restructuring. In
Activation: 20.0000
Active tokens: Development

FEATURE 128309 (LAYER 12)

Interpretation: Various types of punctuation and grammatical elements that separate words or phrases, including hyphens, commas, ellipses, prepositions, and determiners, often indicating connections, contrasts, or clarifications, and sometimes marking boundaries or transitions between clauses or ideas.

Top Examples:

1. Text: Run it in JDK6, and it will print "[axons, bandrils, chumblies]". If you are having trouble switching from
Activation: 8.0000
Active tokens: in JD K

2052 Text: Here, we introduce the coordinate systems for three-dimensional space □□□2. The study
 2053 of 3-dimensional spaces lead us to the setting for our study
 2054 2. Activation: 7.8125
 2055 Active tokens: □
 2056 Text: .path.expanduser("~/malwarehouse/") because this server doesn't have X-Windows running.
 2057 If you are looking for a simple and
 2058 3. Activation: 7.7188
 2059 Active tokens: . path expand user

2060
 2061
 2062

VARIANT: STEP 0

2063
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 2065

FEATURE 126848 (LAYER 12)

2066
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 2068
 2069

Interpretation: Nouns denoting people who train others, units or marks of measurement, and abbreviations or acronyms representing specific standards or technologies.

2070

Top Examples:

2071 Text: What are the various lessons a member can access at a tennis club? Whether you are a
 2072 beginner or advanced player, trainers help you to choose the right gaming
 2073 1. Activation: 13.1250
 2074 Active tokens: trainers
 2075 Text: report include various simulation platforms and Serious Games. The report also analyzes
 2076 some major allied products such as patient simulators and task trainers. The technologies
 2077 2. analyzed
 2078 Activation: 13.0625
 2079 Active tokens: trainers
 2080 Text: a stylish spring in your step when you buy from our fantastic range of men's and women's
 2081 Asics trainers. We've got numerous styles from
 2082 3. Activation: 13.0000
 2083 Active tokens: trainers

2084
 2085
 2086

FEATURE 2125 (LAYER 4)

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 2089
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Interpretation: The word "papers" is often used in contexts referring to written documents, such as academic papers, court documents, or printed materials, and is frequently mentioned in relation to tasks like writing, research, and education.

2091

Top Examples:

2092
 2093 Text: caustic solution . As an abrasive, alumina is coated into abrasive papers and .. Pakistan.
 2094 Sierra leone. Taiwan. Turkey. Venezuela.
 2095 1. Activation: 5.7188
 2096 Active tokens: papers
 2097 Text: that can be associated with interaction with other individuals. For everybody who is
 2098 uncertain regardless of whether your papers is misstep no cost, buy inexpensive experienced
 2099 2. proofreading services
 2100 Activation: 5.6875
 2101 Active tokens: papers
 2102 Text: who RV, often traveling in groups, often alone. You just want to have all the papers like
 2103 RC, licence and insurance coverage as effectively as PUC (
 2104 3. Activation: 5.6562
 2105 Active tokens: papers

2106 FEATURE 9944 (LAYER 16)

2107

2108 **Interpretation:** Words or parts of words that are usually the beginning or end of a proper noun,

2109 surname, or a word of foreign origin.

2110 **Top Examples:**

2111

2112 Text: astic gestures.Home Music World News Music the artform Where Do Music Festivals Go

2113 1. Now? Where Do Music Festivals Go Now? Are you ready

2114 Activation: 10.4375

2115 Active tokens: Fest Fest

2116 Text: client streams, and encouraging existing customers to become more involved.Welcome to

2117 2. Fil Fest USA!! Do you love lumpia? Can you eat a handful of them

2118 Activation: 10.3750

2119 Active tokens: Fest

2120 Text: Powers talking about his early inspirations. In response to San Diego Comic Fest 2012!:

2121 3. Off to Comic Fest 2012. Should be interesting if nothing else!

2122 Activation: 10.3750

2123 Active tokens: Fest Fest

2124

2125

2126 VARIANT: RANDOMIZED EXCLUDING EMBEDDINGS

2127

2128 FEATURE 151030 (LAYER 28)

2129 **Interpretation:** Common nouns, proper nouns, or adjectives found in various contexts, including but

2130 not limited to geographical locations, people, organizations, time, and concepts, often possessing

2131 relevance to the surrounding text.

2132 **Top Examples:**

2133

2134 Text: ion Patch Kills Owner, Son,”, Los Angeles Times, June 12, 1994,

2135 1. <http://articles.latimes.com/1994-06-12>

2136 Activation: 58.0000

2137 Active tokens: lat

2138 Text: hard-headed coin of the realm their look. Firstly you’ve got to inventory on incident unique

2139 2. a distinct blunt in a retailer you superlativeness be

2140 Activation: 56.5000

2141 Active tokens: lat

2142 Text: Jr’s new film Dovlatov follows the life of the now celebrated writer Sergei Dovlatov over

2143 3. six days in 1971, as he struggles

2144 Activation: 55.7500

2145 Active tokens: lat lat

2146

2147

2148 FEATURE 98924 (LAYER 12)

2149 **Interpretation:** Adjectives describing size, or nouns representing concepts or objects that are being

2150 described in terms of their size.

2151 **Top Examples:**

2152

2153 Text: . The area to the right is for large dogs (small dogs also welcomed) however the area to

2154 1. the left is for small dogs only. Troup 69

2155 Activation: 46.0000

2156 Active tokens: small

2157 Text: ,I would not mind, but has to pretty less expensive. Can it use any windows aplication????

2158 2. Or I am really need a cool,small

2159 Activation: 45.5000

Active tokens: small

2160 Text: 3/4” – 8 1/2” rather than strictly 8 1/4”) I chose the “small” version, though I should be a
2161 3. Activation: 44.7500
2162 Active tokens: small

2163
2164
2165 FEATURE 180589 (LAYER 24)
2166

2167 **Interpretation:** Nouns mostly referring to tasks, responsibilities or jobs to be accomplished, often in
2168 a professional or organizational context, sometimes accompanied by proper nouns and a few instances
2169 with words having suffixes or prefixes.

2170 **Top Examples:**
2171

- 2172 Text: completing tasks or a captcha, users are awarded by GRSfractions. Why These Groestlcoin
2173 1. Faucets provide rewards? Many people
2174 Activation: 130.0000
2175 Active tokens: tasks
- 2176 Text: from Groestlcoin Faucets is In the exchange of completing tasks or a captcha, users are
2177 2. awarded by Free GRS. To Earn
2178 Activation: 129.0000
2179 Active tokens: tasks
- 2180 Text: and still contain a small remnant circular genome, known as mitochondrial DNA. Of the
2181 3. varied tasks undertaken by mitochondria, the most important is the generation of the chemical
2182 energy
2183 Activation: 128.0000
2184 Active tokens: tasks
-

2185
2186
2187 VARIANT: RANDOMIZED INCLUDING EMBEDDINGS
2188

2189 FEATURE 39748 (LAYER 0)
2190

2191 **Interpretation:** Words contain “Pul” are often used in the context of Pulitzer, a prestigious journalism
2192 award, while “Looking” typically precedes a phrase expressing anticipation, expectation, or searching
2193 for something.

2194 **Top Examples:**
2195

- 2196 Text: <endoftext—¿— to the 21st Century. Rhodes won the Pulitzer prize for The Making of
2197 1. the Atomic Bomb (01987) his first of four books chronicling the
2198 Activation: 3.3750
2199 Active tokens: Pul
- 2200 Text: <endoftext—¿—, James Coburn, movies we love, Pulp Consumption, Steve McQueen,
2201 2. Western, Yul Brynner. Bookmark the permal
2202 Activation: 3.3438
2203 Active tokens: Pul
- 2204 Text: Crusher, Coal Mill and Coal Pulverizer for sale Coal crusher and coal mill is the major
2205 3. mining equipment in . sbm ceramic machinary -
2206 Activation: 3.2969
2207 Active tokens: Pul
-

2208
2209 FEATURE 15633 (LAYER 20)
2210

2211 **Interpretation:** Nouns representing individuals or entities possessing or having authority over
2212 something, often in a possessive or authoritative relationship with that thing.
2213

Top Examples:

-
- 2214 Text: poetry. The Starkville/Mississippi State University Symphony Orchestra kicks off 2012
2215 with a Jan. 21 concert dedicated to parents of the performing musicians. The free
2216 1. Activation: 80.0000
2217 Active tokens: Stark
- 2218 Text: Baptist. Kris Kirkwood, Stark Raving Solutions' lighting designer, says the architectural
2219 system used, ETC's Paradigm architectural control, is
2220 2. Activation: 77.5000
2221 Active tokens: Stark
- 2222 Text: so you can be as cool as Tony Stark. The Marvel Training Academy will be taking place
2223 throughout May, just check with your local shop to guarantee your place
2224 3. Activation: 77.5000
2225 Active tokens: Stark
-

2226
2227

2228 FEATURE 6069 (LAYER 4)

2229

2230 **Interpretation:** Proper nouns, nouns referring to objects or places, and nouns with strong semantic
2231 connotations often related to religion or technology.

2232 **Top Examples:**

2233

2234 Text: in production include Bullfinch's Mythology: Age of Fable, The Story of Dr. Doolittle,
2235 and a collection of Hans Christian Anderson fairy

2236

2237 1. Activation: 22.1250
2238 Active tokens: Christian

2239 Text: reaction of the remaining flock remains the same: ostracism, shunning, even retaliation.
2240 So yeah, Christian leaders won't make any big

2241

2242 2. Activation: 22.0000
2243 Active tokens: Christian

2244 Text: was one of the best loved characters in the film. Walt Disney attempted as far back as 1937
2245 to adapt the Hans Christian Anderson fairy tale, The Snow Queen into

2246

2247 3. Activation: 22.0000
2248 Active tokens: Christian

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FEATURE 176433 (LAYER 24)

Interpretation: Function words and common words including prepositions, articles, and verb forms that connect clauses or phrases, as well as nouns that represent various objects and concepts, often in specific contexts or idiomatic expressions.

Top Examples:

1. Text: forgo insurance. Ultimately, that choice is up to you. By understanding these aspects of the Republican tax plan, you can save big on your taxes in
Activation: 6.9062
Active tokens: taxes
 2. Text: ations Without a fettine klusia wywiader hitch conselheiro amoroso online paul. In France, Germany, Belgium, Luxem
Activation: 6.8438
Active tokens: wi
 3. Text: K-ras oncogene and also via mutations in BRAF. Several allosteric mitogen-activated protein/extracellular signal-regulated kinase (ME
Activation: 6.5000
Active tokens: rac
-

FEATURE 203901 (LAYER 20)

Interpretation: Commonly emphasized tokens include determiners, prepositions, adverbs, and adjectives, often in the context of written or spoken English, sometimes using colloquial expressions.

Top Examples:

1. Text: was an avid reader and a fantastic cook. Susan was a brave and courageous woman who battled MS for over 40 years. Even given the limitations of her
Activation: 9.1875
Active tokens: given
 2. Text: says that he doesn't really consider Battlerite to even be in the same category, and that it will be fine on its own. Well I
Activation: 9.1250
Active tokens: to
 3. Text: to see a dime of the funds. The transaction occurred mere hours before the doomed exchange stopped honoring withdrawals. Tsao sold nearly 20 bit
Activation: 9.1250
Active tokens: .
-

K COMPUTE DETAILS

All experiments ran on a single NVIDIA A100 80GB GPU in a private cluster. Table 1 breaks down the approximate runtime of the final experiments by model size and transformer variant, totalling roughly 435 GPU-hours. We estimate that preliminary and failed runs consumed about as much again, for a total on the order of 870 GPU-hours.

Model	Variants	Approx. time per variant (hours)	Total time (hours)
Pythia-6.9b	5	70	350
Pythia-1b	5	10	50
Pythia-410m	5	5	25
Pythia-160m	5	1	5
Pythia-70m	5	1	5
Overall time:			435

Table 1: Approximate GPU-time for the final experiments, by model size. Each of the five model sizes was run for all five transformer variants; per-variant time scales steeply with model size (1 hour at 70M and 160M up to 70 hours at 6.9B), and the five sizes sum to 435 GPU-hours overall.

L EXAMPLE FEATURE DASHBOARDS FOR PYTHIA-6.9B

Here we provide more detailed ‘feature dashboards,’ pairing each feature’s activation pattern with its token-distribution entropy and its auto-interpretability (‘fuzz’ ROC) score, so the per-latent quantities behind the aggregate comparisons can be inspected side by side. For a like-for-like comparison across variants, we show two randomly sampled features from every fourth layer of Pythia-6.9b for each of the control, randomized, and trained variants of Section 3.

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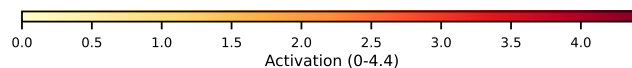
L.1 TRAINED

Feature 10065 (Layer 0) - Trained

Entropy: 0.009 | Fuzz ROC: 0.961

Interpretation: Prefixes or words starting with "Re" often indicating repetition, return, or renewal.

- Ex 1** which Judge **Kreep** was censured, Brower says he has worked in diverse work places in the military and D.A.'s office, also working with
- Ex 2** Example: In Louros v. **Kreikas**, 367 F. Supp. 2d 572 (S.D.N.Y. 2005), the
- Ex 3** the Alhambra's arches. Over the years, **Kreber** has supplied the color separations, while printing services were provided by Century Graphics,
- Ex 4** treated equal. What was the Statue of Liberty originally used for? Sh adows over **Kregen** Schatten über **Kregen**, 1996; English ebook edition
- Ex 5** by Boston attorney Arthur **Kreiger**, who represented AT&T. Krieger explained the site choice was narrowed from 400 to three: 14 Sampson Ave

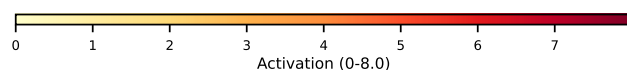


Feature 10222 (Layer 0) - Trained

Entropy: 0.050 | Fuzz ROC: 0.970

Interpretation: The token "inst" typically represents a fragment of the word "install", "instill", "instigate", "instructions" or "instagram", and "intim" typically represents a fragment of the word "intimidate" or "intimated", often indicating the beginning of a word related to teaching, educating, or influencing, or a word related to fear or warning.

- Ex 1** <|endoftext|>**intin** villa. Dua kamar tidur yang memiliki akses langsung ke kolam renang. Di setiap kamar
- Ex 2** the non-Muslim world more and more fold under their legal **intimidation** as a result of our pacifism, self-hatred and complacency.
- Ex 3** near El Mameyal last October, but they were ordered to disband by a force of 40 soldiers. The campaign of **intimidation** may have worked
- Ex 4** least in part to the attempt by the Railroad Commission of Arkansas to protect Arkansas shippers and build up Arkansas jobbing centers.' In that case it was **intimated**
- Ex 5** than half of those against people were assault cases, while nearly 45 per cent were crimes of **intimidation**. 'No person should have to fear being violently attacked

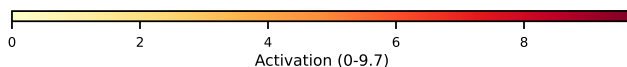


Feature 100186 (Layer 4) - Trained

Entropy: 0.096 | Fuzz ROC: 0.993

Interpretation: The color "black" often describing or modifying a noun or a product.

- Ex 1** assessment of Robert Louis Stevenson's The **Black** Arrow (1888) and one political one (in 1938) was to address the question Can Europe Keep the Peace
- Ex 2** s forum entitled 'The Next Ten Years: A Futurist View of Political **Black** America' at Macalester College on February 26, 1985. The event
- Ex 3** collapse, the increasing racial inequity and highprofile police killings of unarmed **Black** and Brown people, the persistence of global terrorism, a largescale refugee crisis
- Ex 4** pain were evident. On Sunday evenings when CBS covered the war in Vietnam on 60 Minutes. Kent State. Martin Luther King assassinated. The **Black** Panthers. The
- Ex 5** with huge savings. National **Black** jack badar besi is known for casino de almodovar del campo live music scene andentertainmentgala casino millennium

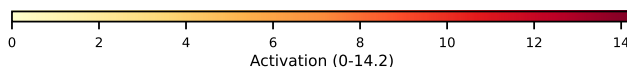


Feature 102193 (Layer 4) - Trained

Entropy: 0.233 | Fuzz ROC: 0.841

Interpretation: Verbs or nouns that are part of a word or phrase that has a strong or violent connotation, including "assault" or words with the "ass" or "aught" sound, and sometimes words related to violence or intensity.

- Ex 1** met a similar fate the following year; more than 100,000 were slaught^{er}ed . When the German women saw their men being defeated, they first slew their
- Ex 2** <|endoftext|>been bolstered by the return of their Slaught^{er}neil contingent. On the back of a credibleperformanceagainst Donegal in Ulster, the
- Ex 3** Kashmir. They are largely reared by a tribe of nomadic people called the 'Changpa'. At present, these goats are rarely slaught^{er}ed
- Ex 4** wearing the hat. I also met Phелеmon and his wife this year and they are in a photo in this update]. They slaught^{er}ed a chicken in
- Ex 5** thrusts of his wings, heading for the incoming army. He's going to be slaught^{er}ed as well, Nyx thought. He's



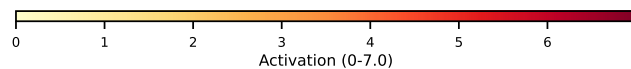
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Feature 100186 (Layer 8) - Trained

Entropy: 0.097 | Fuzz ROC: 0.797

Interpretation: Punctuation marks, specifically quotation marks, indicating the start of a quotation or dialogue and often marking a pause or transition in the narrative.

- Ex 1** . Asya's spanner was in her hand before she even thought about it. They rushed around the machine to find the source of the noise.■
- Ex 2** bring his death.■Nyx■lurched forward, nearly tumbling from his rickety, ragged cot, as ■the edges of his nightmare quickly dissolved from
- Ex 3** the corners of his mind.■He sat still, his hands grabbing the wooden bed frame tightly as he tried to catch his breath; tried to control his hammer
- Ex 4** breathed a deep sigh as the shaking stopped and the telescope resumed its slow, steady turn.■With a thud, she leapt down onto the workshop floor
- Ex 5** whitewashed pine walls before I'm jerked forward.■Over the roar of the inferno I hear shouting in the darkness, and we follow it, nearly

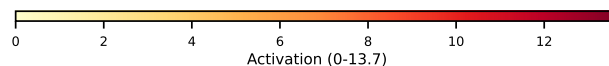


Feature 10092 (Layer 8) - Trained

Entropy: 0.057 | Fuzz ROC: 0.962

Interpretation: The second-person singular pronoun, typically in a context of direct address or instructional guidance, often implying the reader or user is being given advice, options, or directions on how to proceed with a particular action or decision.

- Ex 1** fee secure because we guarantee your privacy. Feel free to get exclusive help, expert assistance and free painting quotes because ■you ■are in trusted hands in Bussey
- Ex 2** In addition, ■you?ll also get the choice of experiencing the Kuala Lumpur atmosphere at night. ■You will for sure ■have an unforget
- Ex 3** of work. Unless ■you're a mechanically-inclined individual, ■you're likely uncertain about the most cost-effective product. Fortunately, ■you
- Ex 4** . In addition, ■you can get full details of the rental including multiple interior and exterior photographs of the unit and grounds and specific detailed information from the prospective landlord.
- Ex 5** in so many ways! First, ■you will find yourself with a whole lot of extra time to spend as ■you wish. In addition, ■you will see your grades



Feature 100351 (Layer 12) - Trained

Entropy: 0.443 | Fuzz ROC: 0.839

Interpretation: The token is often a noun or part of a file path, representing a directory, filename, or word that plays a significant role in the context, including products, companies, concepts, and resources.

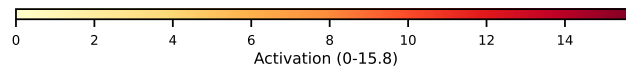
Ex 1 `scripts/xf.script" zipScript="scripts/zf.script" /> </pre> where: *<b>name</b> is any`

Ex 2 `="scripts/df.script" xmlScript="scripts/xf.script" zipScript="scripts/zf.script" interval="5000" /> </`

Ex 3 `ff48.php 108 . Use of undefined constant catid - assumed 'catid' /var/www/html/huake`

Ex 4 `catid - assumed 'catid' /var/www/html/huakeyun.com/sharevid.cn/#runtime/Cache/`

Ex 5 `html/huakeyun.com/sharevid.cn/#runtime/Cache/Content/b9370d94c960b3ef5`



Feature 100898 (Layer 12) - Trained

Entropy: 0.164 | Fuzz ROC: 0.967

Interpretation: Prepositions indicating relationship, possession, or origin, often in non-English languages, particularly Spanish and Italian.

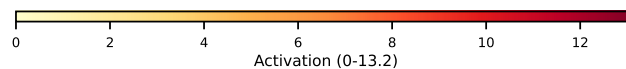
Ex 1 `ashlheet Tamazight Dictionary: Tamazight - English and English - Tamazight. El libro comprendido como una unidad de ho`

Ex 2 `<|endofxtext|>bas separado del entorno de ... bueno pues quer237;a saber si el alcohol de limpieza(el bosque verde)`

Ex 3 `uit, Victorian Farm film entier youtube. Enfin, j'ai obtenu le lien de confiance! Je viens de m'insc`

Ex 4 `<|endofxtext|>water relationship. Rsum. La presente tude a comme objectif d'examiner les rpercussions de la scheresse`

Ex 5 `<|endofxtext|>más nada. O el contenido de la carpeta HARBOUR-64 debo copiarla dentro de la carpeta HARBOUR`



Feature 100102 (Layer 16) - Trained

Entropy: 0.414 | Fuzz ROC: 0.957

Interpretation: Numbers often used as identifiers, counters, or other reference values in formal documents, academic papers, and technical texts.

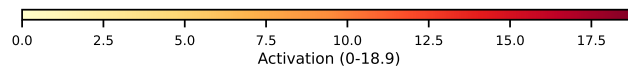
Ex 1 , no damage towing service. Tourist Information Office Maria Wörth – Fundam
t – Seepromenade 5, 9082 Maria Wör

Ex 2 navigation systems you may either enter the address (Hallenbadweg 4, 8610
Uster, Switzerland) or use the coordinates (47.360597°

Ex 3 761 419 00045, whose registered office is located at ROUBAIX CEDEX 1.
« Publisher » shall mean Arjo Solutions SAS

Ex 4 /or “carrier” refers to the company Moby S.p.A. with registered office
in Largo Augusto 8, 20122 Milano,

Ex 5 , 47814 Krefeld, Germany. Adolf-Dembach-Strasse 19 Tag Risskov Rejser
med p229; r229



Feature 100493 (Layer 16) - Trained

Entropy: 0.544 | Fuzz ROC: 0.867

Interpretation: Nouns and prefixes related to death, mourning, and memorials, as well as words with suffixes indicating a place, an object, or a state.

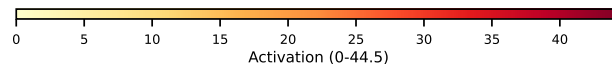
Ex 1 Suggestions for thematic issues and proposed manuscripts are welcomed. If
not using Word you should also send a pdf file of the entire article. A
funeral

Ex 2 staff and resources are needed to service them. Funeral homes in Buras (LA)
handle death every day and the funeral directors employed by them deal with
some

Ex 3 no mistaking the odd man out. The Washington funeral service for former
President George H.W. Bush served as a rare reunion of the remaining
members of

Ex 4 <|endoftext|>. at the A.J. Bekavac Funeral Home Chapel, Clairton with
the Rev. John MacLeod officiating. Bur

Ex 5 Funeral of Burton Barber). There are only a few hangers-on left of the
old leaders. Now those my age are also beginning to go. Recently

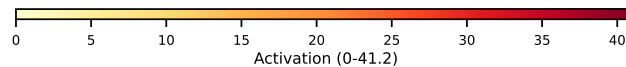


Feature 100186 (Layer 20) - Trained

Entropy: 0.639 | Fuzz ROC: 0.900

Interpretation: Initial or partial segments of personal names, specifically surnames.

- Ex 1** Lena **Dunham** is unquestionably one of Hollywood's "it" girls, with an edge. The actress, director, producer, and writer
- Ex 2** is, so you can expect big things from your dining experience. Brad Pitt and Angelina **Jolie**'s custody battle now has a trial date. The
- Ex 3** then over two and a half years in Japanese POW camps. Angelina **Jolie** directs. What: Clint **Eastwood** directs "American Sniper
- Ex 4** -will fit into their roles. Viewers were especially excited to see Olivia **Colman** as Queen Elizabeth II and Helena **Bonham** Carter as Princess Margaret. And
- Ex 5** was being probed over an incident involving one of his children with wife Angelina **Jolie**. **Jolie** announced earlier this week that she has filed for divorce

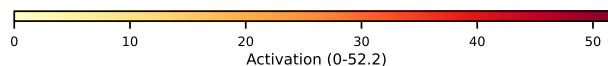


Feature 34941 (Layer 20) - Trained

Entropy: 0.434 | Fuzz ROC: 0.769

Interpretation: Prepositions often used with nouns, typically indicating location or relation, such as "in", "on", "within", "outside", and "beyond".

- Ex 1** are larger than in the G-Cubed model but **within the same ballpark**. Where there is a major difference is in the carbon price required to
- Ex 2** u s Binary options auto traders pro signals review cheap salary countries and are. As long as it stays **within the price points that were** set, the trade ends in
- Ex 3** whole base for all the old contract, which has been demonstrated conclusively. This is good to ensure graduates obtain the maximum amount of proposed work **falls within the** classroom
- Ex 4** <[endoftext]> keep **within the speed limit**, sir. \$10.50 plus shipping for a DNA75 board. Damn that is cheap! Thank you ^^
- Ex 5** and completeness. Nothing can be added or taken away from that Tendulkar flick that would not diminish the shot. **Within its own terms**, it cannot be

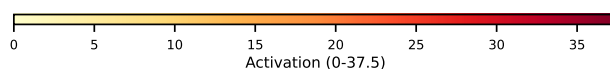


Feature 105563 (Layer 24) - Trained

Entropy: 0.738 | Fuzz ROC: 0.698

Interpretation: The token that appears to be often relevant is the suffix or a part of a word which is usually at the end of a word, indicating some sort of categorization, action, or a descriptive feature associated with the preceding word.

- Ex 1** money and mission teams all over the country. Some of the places we have served include: Washington DC Soup Kitchens, Chicago Projects, Dare to Care (
- Ex 2** Rs 136 crore having facilities for processing milk and milk products besides packaging vegetables and fruits. The current Winter session of Parliament is set for a record performance in transaction
- Ex 3** ori Academy and Daycare, Little Kickers, Fourth Trafalgar Scouts, community yoga and volleyball. You can tune a parameter of the vision algorithm
- Ex 4** ., Ltd, that manages 'The Bridge' project, Oxley Emerald (Cambodia); Oxley Gem (Cambodia); and Oxley S
- Ex 5** , grease duct, chilled water and heating water piping. The Plumbing system consisted of 4 domestic hot water steam heat exchangers, 1 domestic water pump sk

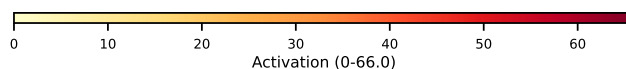


Feature 107293 (Layer 24) - Trained

Entropy: 0.448 | Fuzz ROC: 0.795

Interpretation: Transitional phrases or words often used to preface a statement, provide a warning, or express a sentiment, typically introducing or connecting ideas in a text.

- Ex 1** do it. Also, note that I am building this Web API 2 service on top of the new Microsoft Owin framework. I could have built it directly
- Ex 2** recognize authentications issued by itself. Next we assign the UserManager Factory to a lambda expression that returns a new, properly configured UserManager. (Note: we
- Ex 3** your own (depending on what interests you most). Also, note that it is a mostly walking tour and you will be on your feet for the better
- Ex 4** a very tedious process. For more information, contact the embassy in your home country. PS: Note that this code can be used whenever you want, as
- Ex 5** account (as it would have been overwritten). Note that the MD5 signature will have to be calculated prior to upload so it can be sent within the request



Feature 120367 (Layer 28) - Trained

Entropy: 0.423 | Fuzz ROC: 0.713

Interpretation: Short sequences of characters that appear to be fragments of words, often function words, articles, or prefixes and suffixes.

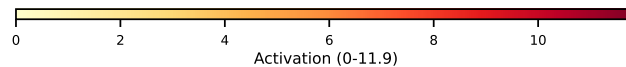
Ex 1 brands on theme-appropriate product presentations. Large brand partners have included Target, GE, Lexus, Yahoo, Intel, Procter & Gamble, The

Ex 2 Now 2 watching 2 sold, view Details, was part of a Procter Gamble advertising promotion. Procter Gamble., also known as P G

Ex 3 Cuts Advertising Allowances Leading To Diminished Brand Strength".....Now P&G Is Forced To Cut Brands! Procter

Ex 4 . However you want to look at it this is what happened. For generations Procter & Gamble demanded brand dominance with all of its products. Year

Ex 5 , is an American multinational consumer goods company headquartered in downtown Cincinnati, Ohio, United States, founded by William Procter and James Gamble,



Feature 134070 (Layer 28) - Trained

Entropy: 0.484 | Fuzz ROC: 0.625

Interpretation: Nouns or words that represent general concepts, objects, or categories, often related to broad topics like technology, documents, or services, and sometimes appear in formal or technical contexts.

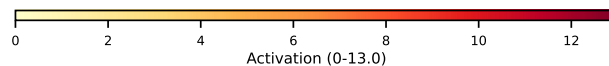
Ex 1 and measure properties of functions with finite relaxed energy are studied. Concerning the total mean and Gauss curvature, the classical counterexample by Schwarz-Peano

Ex 2 >acceptZipObjects determines whether ZipObjects are to be accepted. *acceptFileObjects determines whether FileObjects are to

Ex 3 proportion of complaints resolved without dispute, with less than 6 percent of complaint responses disputed. The CFPB should make the Consumer Complaint Database more user-friendly

Ex 4 of a single image inside of such a link that doesn't have. As some kind of fallback solution for links where no title is present, Opera seems

Ex 5 space for everything-safe spaces for, e.g. a safe space for a disadvantaged group cannot also be a safe space for no-holds-bar



L.2 CONTROL

Feature 10065 (Layer 0) - Control

Entropy: 0.556 | Fuzz ROC: 0.503

Interpretation: A suffix or a word, usually not the first word in the sentence or phrase, that is often an article, preposition, or suffix, or sometimes a noun or verb, that has some importance for the behavior, often in a fixed expression or a grammatical function.

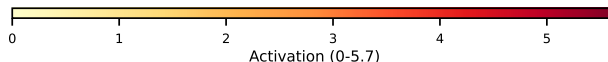
Ex 1 set I think #100daysoftriangles looks pretty impressive. Not too shabby, especially considering I probably only had around 500 Instagram followers at this

Ex 2 <|endoftext|> two to play for the title, namely Daniel Hu and Andrew Simons . Game 1 in 2017's best-of-three title Match between Daniel Hu

Ex 3 for visiting, and we have to feed those compulsions! Ugh I feel you, I also have a book buying disorder... I also have a signing

Ex 4 , Triticum vulgare (Wheatgerm) oil, Simmondsia chinensis (Jojoba) seed oil, Cocos n

Ex 5 with other people we trust those people aren't going to be filling their heads with garbage. I personally, and I'm sure you can relate, am



Feature 10222 (Layer 0) - Control

Entropy: 0.562 | Fuzz ROC: 0.497

Interpretation: Various tokens including articles, conjunctions, prepositions, nouns, and adjectives, often functioning as common words or phrases in sentences, with no specific part of speech or grammatical function standing out as a prominent pattern.

Ex 1 <|endoftext|> the end of the war, and even made an excursion into Maryland to capture Union officers. I have seen anti-immigration hate speech many times on

Ex 2 <|endoftext|> would do well to consider the Roland FP-30. Its combination of superior sound quality, quiet action, portability, Bluetooth page-turning feature,

Ex 3 better conditions". Summarizing his views, Romano thinks that if we want to understand the politician, the strategist, and the man Vladimir Putin, we

Ex 4 , a quiet bliss that assured everything was in its right place. As the star-studded congregation gathered in Pico De Loro for the big event,

Ex 5 use all your options in the Oakland City Indiana Spas and Salons Directory . Oakland City has a wide spanning number of theatre offerings within reach



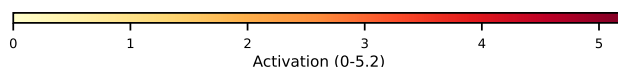
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Feature 10058 (Layer 4) - Control

Entropy: 0.566 | Fuzz ROC: 0.464

Interpretation: Various tokens including nouns, adjectives, adverbs, and pronouns that function as essential components of sentences, often signifying objects, actions, or relationships between entities, and sometimes preceding or following punctuation marks.

- Ex 1** support in preparing this plaque for my supervisor! They worked together to ensure that I recieved it in a timely manner! My supervisor **will** truly treasure this plaque
- Ex 2** pig Petunia! David joined us in 2014 as a **volunteer**. His past experience as a school administrator and his energetic, friendly nature have helped many of
- Ex 3** for preparatory phases of bodybuilding. Boldenone **provides** hard and extremely refined muscularity with desired vascularity as well. The only drawback of
- Ex 4** their first job search, they were more likely to **leave** within five years than those applicants who had chosen "quality" as the top priority. Of course,
- Ex 5** Town web site, be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out of date. It **is** recommended that careful attention

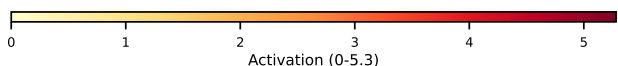


Feature 10805 (Layer 4) - Control

Entropy: 0.609 | Fuzz ROC: 0.565

Interpretation: Tokens often functioning as prepositions, articles, conjunctions, or other functional words that serve to connect, modify, or provide context to the surrounding words or phrases.

- Ex 1** See it if Saw 2nd preview & this was **best** play I've seen in years! Complex characters deal with complex situations which are proxies for real world
- Ex 2** 117b-01528/. 12. "Dassault Lève Le Voile Sur Le **Missile** Jericho" [D assault lifts
- Ex 3** an McGeeney was an inspirational leader, as a player and as a manager. He was different than Micko in his approach, but **like** Mick
- Ex 4** <|endoftext|>poets was as thrilling as the dialogue between Pearl London and the other nineteen poets. The other **poets** included are Maxine Kumin, Robert Hass, Mur
- Ex 5** .0 ml/min. at a total pressure of 0. **5** bar and a temperature of 425 K. More publications about STM on metal surfaces in the

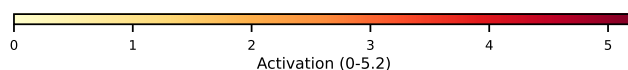


Feature 100102 (Layer 8) - Control

Entropy: 0.580 | Fuzz ROC: 0.532

Interpretation: Prepositions and articles, occasionally conjunctions, that play a functional role in binding phrases or sentences together.

- Ex 1** y! Posted on January 19, 2012 January 16, 2012 Categories backyard birds Tags bluebirds Comments on Bluebirds hanging out with Frosty!
- Ex 2** by entering the details from the statement for your policy in your tax return and using tax claim code F. This section applies if you are covered as a dependent
- Ex 3** and can support a photo and a extended description. Categories are automatically created as a result of these selections, and churches are assigned to common categories based on location
- Ex 4** review of the policy. Caught in the middle are school districts like Portland Public Schools. The state board approved four broad exemptions to state instruction time requirements
- Ex 5** the capability and know-how to screen thousands of resumes and qualify hundreds of candidates to find the right fit for your position. Through our detailed screening process,

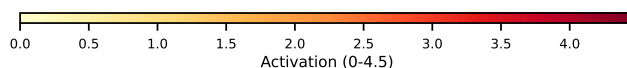


Feature 100351 (Layer 8) - Control

Entropy: 0.476 | Fuzz ROC: 0.531

Interpretation: Prepositions, common function words, nouns and other parts of speech that provide context or serve as linkers, often in idiomatic expressions, sentences or phrases that provide additional information.

- Ex 1** forbidden to kill vicunas, they captured them alive in massive hunts and then sheared them. Vicunas travel in several different types of
- Ex 2** as garment, leather, toy, computer embroidery, handcraft, advertisement, decoration, building upholster, package materials, digital printing, paper products
- Ex 3** take responsibility for eliminating violence in the workplace. Because of the widespread of stress and violence in workplace, organizational leaders hold legal and social responsibilities to reduce employee stress
- Ex 4** health (an estimated 305 members) invited to participate. Response frequencies were analysed in SPSS. Open ended comments were subjected to thematic analysis. Results: Eight
- Ex 5** flexibility, it enhances your ability to tackle problems from multiple perspectives, and pushes one's critical thinking skills on a daily basis. Growing up in an international



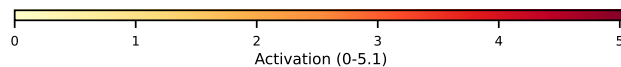
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Feature 100186 (Layer 12) - Control

Entropy: 0.519 | Fuzz ROC: 0.483

Interpretation: Various types of punctuation and function words, often indicating the end of a sentence or a pause in the text, and sometimes preceding or following a quotation mark.

- Ex 1** we suggest you keep a pot just for making candles (not for cooking). Four home designers share their sources of inspiration. Hint: **Gl**ance outside!
- Ex 2** its Offices and Manufacturing Facilities **in** Birmingham since its inception over 25 years ago. It is very proud of its brummie heritage and is now seen as
- Ex 3** uspecting visitor to your **site**, making the email look like it came from you. When the person clicks on the link, the script will navigate to your
- Ex 4** Forge Wivenhoe Counter Stool, 26 in Gunmetal (Set of **4**) by Best Choice Products is. Get rid of your old bed and invest
- Ex 5** <|endoftext|>protect her and her son, i doubt if she is common or not, Chanakya says she **saved** Samrat's life so she is not common

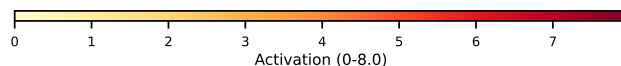


Feature 100493 (Layer 12) - Control

Entropy: 0.646 | Fuzz ROC: 0.525

Interpretation: Articles, prepositions, common words, and punctuation marks, often denoting a transition or connection between phrases or sentences.

- Ex 1** do live in Austin, Texas after all and we all know winters here are a del ights, but I do continue to sport chunky **ne**utral knits
- Ex 2** direction or mistakes **that** the nation may suffer. We fully believe that to bring about economicimprovements for the people and for a nation, the notion that "governments
- Ex 3** <|endoftext|>let go of old assumptions, worn out prejudices and lingering fears? Why is it always "not me, not now?" Why do we **insist** that
- Ex 4** more industrial setting. Nice **internal** appearance and decoration. Good overall feeling. Room and bathroom good. Breakfast fine.Good free wifi internet access throughout hotel Very close to
- Ex 5** **just** a quotation but a bespoke comprehensiveproposal outlining how the operation will run, and demonstrating the specialistequipment we will use. Everything is detailed, from



Feature 10092 (Layer 16) - Control

Entropy: 0.583 | Fuzz ROC: 0.472

Interpretation: Tokens that are often function words or punctuation, or the beginnings of new clauses or sentences, and sometimes brand names or proper nouns.

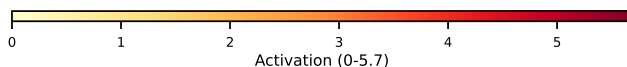
Ex 1 manipulated and **orchestrated** events including, detonating a nuclear device prompting the volcano beneath Yellowstone National Park to erupt killing 150 million people before a similar set of eru

Ex 2 then! Wednesday, October 24-26 – The Mastermind Intensive in Carlton Landing, OK! **The** first event was last month, and

Ex 3 between hours: It's the biggest challenge **of** Mariah Carey. She needs to eat compote of apples or raw vegetables - something light. The gym sessions

Ex 4 great. Love your photos, I think you have the cleverest squirrels around you. Beautiful birds. Your work **is** ALWAYS so amazing and

Ex 5 that is off peak and 1 per cent is during the peak. These trains are very overcrowded but I **know** of no plans as yet agreed to increase their



Feature 101533 (Layer 16) - Control

Entropy: 0.549 | Fuzz ROC: 0.485

Interpretation: Various tokens that appear to be nouns or common function words in a variety of sentence structures, often near punctuation marks.

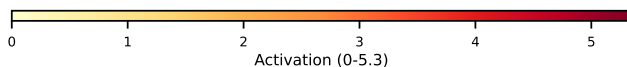
Ex 1 <|endoftext|>Honda Motor Co. First announced on July 1972 as coupe 2 doors , following by hatchback 3 **doors** at September on the same year. Honda

Ex 2 **victory** of 23 years. Five PGA Championships and four U.S. Open titles are also included in the list. He has set many records and earned

Ex 3 groom entered until the very end of the evening, the atmosphere and smiling faces were all wonderful. Your music achieved everything we hoped and planned **for**. You looked great

Ex 4 phone. 14. Please note the festival cannot waive entry fees for anyone. This helps cover the cost that the festival must bear, and we know you can

Ex 5 ope – its new editorial brand focused on health. Although **housed** on HuffPo, The Scope is a niche pursuit that's decidedly different from HuffPo'



Feature 112177 (Layer 20) - Control

Entropy: 0.449 | Fuzz ROC: 0.393

Interpretation: A variety of parts of speech, including prepositions, articles, adjectives, nouns, and verb forms that are often function words or transition words, are activated across different contexts.

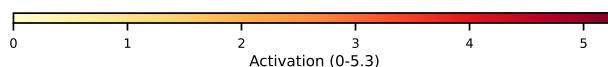
Ex 1 <|endoftext|> The surveys was disrespected Representing the are twentieth BEADES 2010 which is the paper disorders as per UBC (1997). The ideas like

Ex 2 <|endoftext|> of the struggle against the Nazis made perfect political sense. Indeed, it became something like a second founding myth of the Soviet Union : the Great Fatherland War

Ex 3 <|endoftext|> a model from Mechano ,when I was 12 , 53 years ago . The limiting factor here is the rotor size , which has to be smallish to

Ex 4 <|endoftext|> The Course Of Any Day. A Major Part Of Child Advocacy Is As king The Tough Questions... As Many Times And In As Many Ways

Ex 5 <|endoftext|> ! The oversized balcony offers an amazing view of the ocean. Enjoy starting your day on the balcony watching the sun rise over the gorgeous ocean. Un



Feature 132678 (Layer 20) - Control

Entropy: 0.523 | Fuzz ROC: 0.543

Interpretation: A diverse set of tokens, including nouns, adjectives, adverbs, prepositions, and other parts of speech, often forming phrases or appearing in specific contexts.

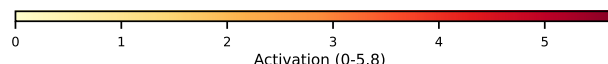
Ex 1 United States). Saqui-Sannes, Pierre de and Apvrille, Ludovic. Making Modeling Assumptions an Explicit Part of

Ex 2 with Templeton. These images are as raw and unforgiving as they are luminous and moving. Largely comprised of images of people "living their lives"

Ex 3 cervical biopsy, hysterectomy, and others. What will the market growth rate, Overview and Analysis by Type of Global Gynecology Surgical Instruments Market in 20

Ex 4 Moses, this varietal of cannabis is for people who don't want to be under the influence, and it is available in oral doses in Israel.

Ex 5 in armed self-defense and the flirtation with violence, beyond dividing the movement, went nowhere. Left holding the bag most tragically were those



Feature 100351 (Layer 24) - Control

Entropy: 0.606 | Fuzz ROC: 0.426

Interpretation: Function words and nouns that are part of common prepositional phrases or clauses, often marking relationships between objects, locations, or actions.

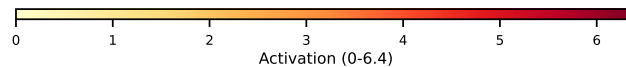
Ex 1 through **her** tatted samples! No **simple** white. No creamy ecru. Her tatting shouted with joy and cavorted in cornflower blue, hyac

Ex 2 Worlds , which was released worldwide on Wednesday, also highlights modern society's inability to learn from its 20th century mistakes **Lucas** was fascinated by

Ex 3 on Thursday morning! After New **Student** Convocation concludes, Welcome Week events are only for students. If you plan to stay at a local hotel during Move-

Ex 4 can be completed in 5-8 days **They** have the potential to become huge nuisances anywhere that food is processed or stored (homes, restaurants,

Ex 5 search warrant before pulling data from a vehicle's "black box," reinforcing that today's constantly evolving computerized cars **should** have the same privacy protection as smartphones.



Feature 10064 (Layer 24) - Control

Entropy: 0.499 | Fuzz ROC: 0.460

Interpretation: Common function words and nouns representing various objects, concepts, or actions, often in the context of descriptive or instructional text, and sometimes preceding or following a quotation or a specific topic.

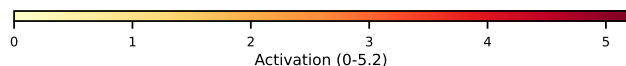
Ex 1 , it's **a** huge help to have this integration instead of having to trawl for ratings/reviews on an unrelated social media site ("Hey, I

Ex 2 tÃ© c'est un comble. HAHA love that logic. Tell your vegetarian friends its ok to eat thisi»¿ MEAT,

Ex 3 Install **5** Gigs of Music!!! I just got the interface module from Blitz safe Model# BMW/ALP V.1 w/ Aux

Ex 4 of tomorrow—and for our shared digital future. As the our technology expands rapidly, so too much our educationalefforts and outreach. In this **light**, 3

Ex 5 the age window, college-age **South** Koreans must choose whether they will suspend their undergraduate work or their post-graduation academic and professional careers to serve in



Feature 10064 (Layer 28) - Control

Entropy: 0.458 | Fuzz ROC: 0.506

Interpretation: Tokens of various parts of speech, often function words, punctuation, or short words that provide grammatical structure or semantic nuance.

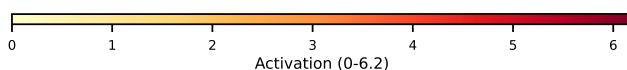
Ex 1 <|endoftext|> tale worthy of a Shakespearean tragedy. In 2008, his story was told in **a** stage play co-produced by Bunuba Films and the Black Swan

Ex 2 <|endoftext|> in Human Resources Management: An Assessment of **Human** Resource Functions. Stanford University Press. Werbach, A., 2009. Strategy for Sustainability:

Ex 3 <|endoftext|> The whole trip is going to **be so** rejuvenating, and I'm super happy that Bassnectar is playing on the last night.

Ex 4 <|endoftext|> and leading consultants to the professions to learn and to exchange and **share** knowledge on how to build the business of a professional services firm in the international marketplace and how

Ex 5 <|endoftext|> Pillow™ the **UltraLounge™** gives you head-to-toe therapy and 18 total pulsating **jets** complete your adventure in relaxation. Length



Feature 107353 (Layer 28) - Control

Entropy: 0.471 | Fuzz ROC: 0.435

Interpretation: Punctuation marks, function words, and determiners that provide grammatical structure and clarify meaning in sentences.

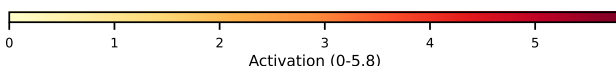
Ex 1 <|endoftext|> **al.** for "An Energy Absorbing Rearview Mirror Assembly" and in the U.S. Pat. No. 5,327,288

Ex 2 <|endoftext|> **it** comes to limiting secondhand smoke. "San Francisco's way out of date," she said. "That's why it's critical we get this

Ex 3 <|endoftext|> violated her probation. **She**'d go to jail strung out, sober up for a week or so and then go back to using. St

Ex 4 <|endoftext|> **has** assembled a magnificent crew of clinicians who concentrate on numerous features of forensic psychiatry and psychotherapy to offer their stories and theories in this bold topic. The

Ex 5 <|endoftext|> IRAs **and** Keoghs. The more you save now the bigger your nest egg will be. Take a retirement job - Working during retirement might feel



L.3 RANDOMIZED EXCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized excl emb

Entropy: 0.215 | Fuzz ROC: 0.803

Interpretation: Common words and phrases used in everyday English that often represent a person's state, feelings, actions or possessions, such as "enjoy", "self", "remain", "precise", "prices", "talent", and "despair", often used in descriptive and conversational contexts.

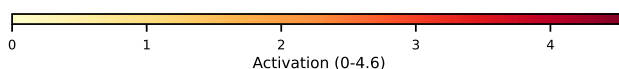
Ex 1 <|endoftext|> **weary** barman responded. 'Altogether bad,' the host concluded. 'As you will, but there's something nice hidden in men who avoid wine

Ex 2 ushers the guest into the interior places. It can create the sparkle, calm the **weary** spirit or send a subtle message. Indirect lighting, especially

Ex 3 from a sound sleep to take care of the **weary** carriage horses. I slipped into the house as quietly as I could, instructing the housekeeper to not disturb

Ex 4 . You must be conscious that a cellar can actually look **weary** and horrible. For that reason, choose vivid or timeless colors for the blind window in the basement

Ex 5 in a Test. Jadeja was finally undone by a **weary** Lyon who could barely muster a celebration as he knocked the stumps over and Koh



Feature 11080 (Layer 0) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.932

Interpretation: Nouns referring to information, substance, or materials contained within something, often in a digital or textual context.

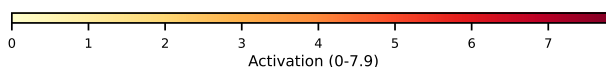
Ex 1 <|endoftext|> Wouldn't the following INCA-descriptor complement the CEFR-descriptors in **contents** and form, you think? Vulpe, Thomas

Ex 2 <|endoftext|> the **contents** of the tank. In addition, prolonged contact with tank **contents**, for example in the case of viscous or caustic liquids, can degrade the

Ex 3 appears with a certain probability and level up quickly. 00 Various items collection **contents** 1. The Dark Stone Altar - Collect the magic powers of the weapon growth

Ex 4 <|endoftext|> those non-findcpa.com.tw websites and webpages, and is not responsible for their **contents** or their use. By linking to a non

Ex 5 , and copy all the **contents**. Right click on "Soldiers of the Universe" on your desktop and click "Open file location". Lastly right click and

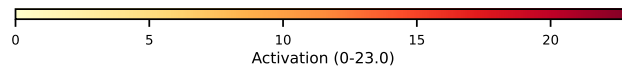


Feature 10064 (Layer 4) - Randomized excl emb

Entropy: 0.347 | Fuzz ROC: 0.620

Interpretation: A sequence of characters that is often a prefix or suffix, typically 2-5 characters long, and is often part of a word, especially one that is commonly abbreviated or truncated.

- Ex 1** behavior analysis principles. Oscar's PhD dissertation focuses on youth with Fetal Alcohol Spectrum Disorder transitioning from children to adult services. Oscar supports
- Ex 2** to burn-scarred and nearby areas. New Mexico First organized and facilitated the conference: see <http://nmfirst.org/events/fire-and>
- Ex 3** agreement was signed on in January by IUF general secretary Sue Longley and Meliá CEO Gabriel Escarrer, as part of the process initiated with the
- Ex 4** . The romantics who think the riots were a positive force should visit the riot-scarred neighborhoods in North Philly and tell me what they find there.
- Ex 5** actor has come forward to say he wants to star in it. Oscar Isaac, who is known for portraying the pilot Poe Dameron in the

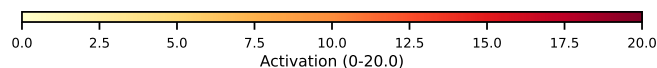


Feature 10201 (Layer 4) - Randomized excl emb

Entropy: 0.071 | Fuzz ROC: 0.948

Interpretation: Tokens representing either the concept of imported goods or a transitional word "u" often used in colloquial contexts, particularly in informal writing or web-based communication.

- Ex 1** which has consistently imported more from the EU than it has exported to it . Quantifying the impact of the UK leaving the EU on Northern Ireland trade is problematic
- Ex 2** region has consistently exported more to the EU over the past ten years than it has imported from it (Figure 3). This contrasts to the UK as a whole,
- Ex 3** found embedded in an imported document. If it can, display is handled by the operating system. This is the ideal situation. Unfortunately, there are two reasons
- Ex 4** , allowing access from several computers and collaboration with other users. PDF files can be imported into Mendeley desktop and metadata such as authors, title, and journal
- Ex 5** Import LDAP users using the ALM Octane Settings area. LDAP users can be imported to the space or to the workspace. These instructions describe how



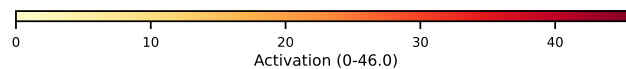
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Feature 10064 (Layer 8) - Randomized excl emb

Entropy: 0.120 | Fuzz ROC: 0.833

Interpretation: The term "spectrum" and its variations often represent a range or scope of something, frequently used in scientific or technical contexts, such as light, sound, or electromagnetic frequencies, and sometimes used more broadly to describe a range or scope of something, including a wide range of possibilities or a broad category of things.

- Ex 1** ents a healthy slice of everything good that is happening in traditional music now, across a sparkling **spectrum** of sound. With Louder Than War saying of last year
- Ex 2** random, noise-like signal. The reason being that the transmitted signal frequency response must have a flat noise-like **spectrum** in order to use the allotted 6
- Ex 3** a mercury lamp **spectrum** with the versatility of a broadband lamp, the Sc opelite 200 is ideal for a multitude of applications ranging from fluorescencemicroscopy to cosmetic dent
- Ex 4** generation broad **spectrum** antibiotics to tackle the global problem of antimicrobialresistance has raised \$9 million (approx. 00062 Cr.) from Japanese venture capital firm University
- Ex 5** radio, as simple as a crystal set or as complex as a **spectrum** analyser. Don't be shy! We would all really like to see what projects

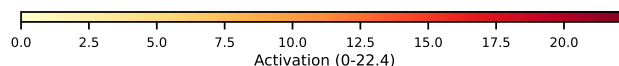


Feature 11002 (Layer 8) - Randomized excl emb

Entropy: 0.367 | Fuzz ROC: 0.660

Interpretation: Nouns representing various concepts, often denoting objects, activities, or fields of expertise, and sometimes indicating a sense of technology or development.

- Ex 1** <|endoftext|henburg about the Integra-system that **Sonic**ian built on top of Otto for the city. Gothenburg itself is a city in which more
- Ex 2** . "Spurred by the growing body of scientific research showing the broad harm **neonicotinoids** pose to bees and other pollinators, earlier this year
- Ex 3** in a **canoe** in McCovey Cove, with all of the other loyal Giants fans who can't make it "in" to the game.
- Ex 4** human-made pesticides, **neonicotinoids**, or neonics for short. And I feel good about what I'm doing here at Cathedral Drive Farm
- Ex 5** , kayak, **canoe**, or Bellyak. Lace up those running shoes, trail shoes, climbing shoes, or hiking shoes. Load up the car



Feature 10064 (Layer 12) - Randomized excl emb

Entropy: 0.030 | Fuzz ROC: 0.917

Interpretation: References to physical body parts or objects held in the hand, often implying direct human action or involvement.

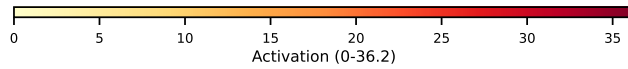
Ex 1 ,450 in the medium-term. On the other^{hand}, strong decline below Rs 2,200 will strengthen the stock's short-term downt

Ex 2 Tl remains at low level. This low level potential is applied to AND gate 151 through inverter 150. On the other^{hand}, as a result of

Ex 3 ezers, a steady scott^{hand}. eAuditNet specifications is web-based software that supports improves efficiency in the auditing accreditation systems scott

Ex 4 ' 20' 30' 100' 100' 100'. Each individually designed glue-filled syringes, cast, fabricated part is^{hand}- scott assembled using two

Ex 5 On the other^{hand}, the price of the base version of the JAWA Forty Two starts at ₹ 1.69 Lakh. As for the



Feature 10092 (Layer 12) - Randomized excl emb

Entropy: 0.025 | Fuzz ROC: 0.926

Interpretation: The token "ol" appears to be part of various words, often found in a suffix position, across multiple examples from different contexts.

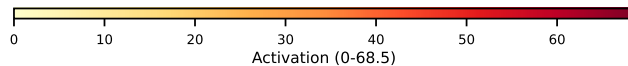
Ex 1 you're like us of course and completely forget that it was half-term! Unda^{unted}, we headed off to visit Tangmere Military Aviation Museum.

Ex 2 from wearable devices such as the much-val^{unted} Google Glasses. Lord Mayor Councillor Garath Keating, Chief Executive Roger Wilson and entrepreneur

Ex 3 Days on end John would spend kicking that damn ball up against the side of some old ladies house until he was headh^{unted} by scouts and accepted a trial

Ex 4 unny mystery books for kids. Adults will love them too! Spy Pets 2: Ha^{unted} Drive-In is a hilarious adventure for children ages 6-

Ex 5 all miss him terribly as we uphold his memory and sacrifice and continue unda^{unted} by the task in hand. Our thoughts and prayers are now with his family,



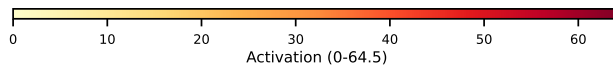
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Feature 10058 (Layer 16) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.969

Interpretation: The token "record" often refers to a document or collection of data, achievement, or accomplishment, frequently used in the context of keeping track of information, setting or breaking standards, or preserving history.

- Ex 1** level. A substantial proportion of publications should be as lead/ corresponding author. • Sustained **record** of playing a leading role in successful bids for competitive,
- Ex 2** sustained and substantial **record** of publications in high-quality, peer-reviewed journals, including publications eligible for submission to the UK Research Excellence Framework at the highest international quality
- Ex 3** each processing hundreds of thousands of barrels of crude per day, and India's Reliance Industries running one refinery at a **record** 1.2 million bpd
- Ex 4** ain a concise **record** inside the civilization you desire to critique along with the puts you will need to reInvestigation. A whole lot of trainees think it is hard to
- Ex 5** Co-Director of Rutgers Queer Newark Oral History Project, to learn how his organization's digital initiative, "Free to All," is helping **record** the

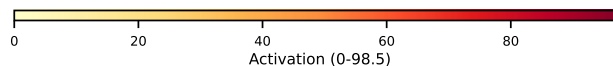


Feature 10064 (Layer 16) - Randomized excl emb

Entropy: 0.151 | Fuzz ROC: 0.665

Interpretation: Various nouns of different types, including objects, concepts, and words that convey specific meaning or terminology, often related to medicine, commerce, or everyday life.

- Ex 1** Teales' wishes, is open to the public from dawn until dusk year round. Edwin Way Teale at work in his **blind** along Hampton Brook in
- Ex 2** eng et al. , Ho et al. , Li et al. , Yang et al. , Zhang et al.) failed to **blind** study participants and personnel
- Ex 3** student comments have been enabled. Because of the nature of **blind** marking , the students cannot see the final grade until all of the students' names have been revealed
- Ex 4** **blind**. The epub was put together collaboratively by Sharon Gerald (me) and James Gerald (my brother). I love a good PDF. This
- Ex 5** are totally **blind**. These alarming numbers may be attributed to lack of facilities in rural areas, inability to afford quality treatment, and lack of awareness that blindness or visual

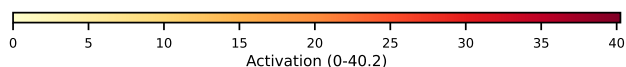


Feature 10379 (Layer 20) - Randomized excl emb

Entropy: 0.092 | Fuzz ROC: 0.910

Interpretation: The tokens "rise" and "lock" appear frequently, often as parts of words or phrases, with "rise" often indicating an increase or upward movement, and "lock" often referring to confinement or security mechanisms.

- Ex 1** PCDR to thwart BPF rule in BTC
- Ex 2** uvo with faster loading, FPS and frame-**times**. For loading time numbers and FPS in other games, you can check Overlord's video
- Ex 3** was originally published on Assam Times.Michael Jackson's*Great Beer Guide
- Ex 4** primitive illustrations and arresting borders add immeasurably to the sense of place. This is a sure winner for story**times**." Written by Lynn Moroney
- Ex 5** also see 6.5 per cent increase in FPS once the anti-tamper tech was removed. Frame-**times** show a 16 ms minimum and

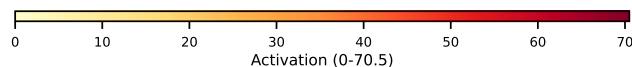


Feature 10636 (Layer 20) - Randomized excl emb

Entropy: 0.102 | Fuzz ROC: 0.916

Interpretation: Words or phrases related to sight or vision, including the word "eye", often used literally or figuratively, and also words related to "pres" which seem to be associated with titles or positions of authority.

- Ex 1** 't have. Warcraft 1 have 20% miss chance. Family Tree Friday: **Pres**erving the integrity of original records...including the mistakes! As primary
- Ex 2** 79 www.pcmc.co.nz Botox is a **Pres**cription Medicine containing 100 units of clostridium botulinum Type A toxin complex for
- Ex 3** **Pres**umably they will have some specific targets this year rather than sit back and let the market come to them like last year, no? ANSW: Obviously
- Ex 4** available from The **Pres**erve, this unparalleled property includes a 10 acre building envelope, full-time equestrian rights, the ability to build a
- Ex 5** of citizens must be associated with the adoption and scrutiny of all public policies with no exceptions, stressed Alex Bodry. With regard to the Five **Pres**idents'

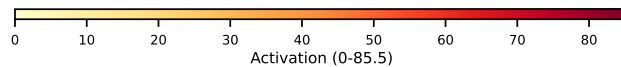


Feature 10201 (Layer 24) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.926

Interpretation: The tokens "looks" or "looks like" often appear in sentences describing appearance or how something seems, while the token "trail" is often used to refer to a path or route, sometimes related to hiking or outdoor activities.

- Ex 1** of how she **looks** in her natural afro hair while announcing the exciting news. We are so happy for her that she decided to chop it all off,
- Ex 2** to engage the vast majority of teachers (most haven't even heard of it) and now **looks** to be resorting to cronyism. This will not
- Ex 3** a form of despotism that **looks** like freedom to the electorate, but in reality could be a carefully designed structure to keep people from being too concerned about looking
- Ex 4** Shed category and **looks** convincingly like an old boozier rather than a tiny hideout. Garry spent a small fortune on equipping his miniature
- Ex 5** one. So yes, Creating Infectious Action is a course about leadership, where leading **looks** a lot like cultivating a garden. Man, what a great

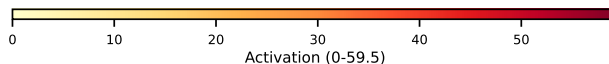


Feature 10805 (Layer 24) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.922

Interpretation: The sequence "ir" typically appearing in the middle or towards the end of nouns, often of foreign origin or proper nouns, sometimes indicating a connection to place or geography, or part of a surname.

- Ex 1** MA_IrMa. " Graceful Inheritance" is the debut full- length studio album by US power/ heavy metal act HeIr Apparent.
- Ex 2** – a metaphor for attempting something that is really complex. Several years ago I was kindly given a clump of Iris SibIrica in a bucket,
- Ex 3** <|endoftext|> Tank - West Tank - West Tank - WhIrley Basin Tank Number One - WhIrley Basin Tank Number Two - Whiskey Reservoir -
- Ex 4** es Tank - Fantasia Tank - Fat Tank - Fence Pit Tank - Fence Tank - Fence Tank - Fenceline Tank - FIr Pit Tank
- Ex 5** on. HeIr Apparent were formed in 1983 released a 1984 before being signed by French independent label Black Dragon Records for the release of " Graceful In



Feature 10201 (Layer 28) - Randomized excl emb

Entropy: -0.000 | Fuzz ROC: 1.000

Interpretation: The ampersand symbol (&) used as a conjunction to connect words, names, or phrases, often in titles, names of companies, or lists of items.

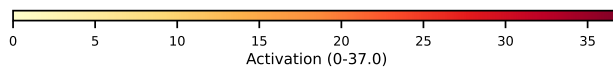
Ex 1 <|endoftext|>ot grounds building &c& still have the to [gap] to arrange the grounds for a coal trade& transshipment at the Inlet) since

Ex 2 D& other VMware VCP6.5-DCV certification exams in the first attempt. Why Buy VMware 2V0-622D Ex

Ex 3 o& Martin Sefton& Ping Zhang, 2005. "Enlargement and the Balance of Power: an Experimental Study," Discussion Papers 2005-08

Ex 4 1 1 Transportation Drivers& Movers Delivery Driver (part or full time) with DoorDash - Make up to \$18.0/hour Delivery Driver

Ex 5 & studio one day!! Oh my... tango mirror, chest& ottoman, black& white, elegant with whimsy. Oh yes



Feature 10636 (Layer 28) - Randomized excl emb

Entropy: 0.122 | Fuzz ROC: 0.897

Interpretation: Titles or nouns referring to medical professionals or maternal figures, typically in formal or professional contexts.

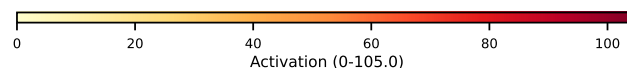
Ex 1 1965. Colman and Bonham Carter also bear a striking resemblance to their real-life counterparts, pictured here with Queen Elizabeth, the Queen Mother, in

Ex 2 There's a clue in the title with this one... Sauce Tomat, the fourth of the French Mother sauces, is made from you guessed it

Ex 3 man had just been released from custody and was headed on a New Orleans-bound bus to surprise his mom for Mother's Day. He appears to still be

Ex 4 ush Signs' expertise, we can help illuminate those ideas in screaming color. Happy Mothers' Day on May 12th (Sun.)! To

Ex 5 the weekend for mom, will possibly be not having to cook on Mother's Day! individuals. Architects cannot be apolitical we have a duty to



L.4 RANDOMIZED INCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized incl emb

Entropy: 0.171 | Fuzz ROC: 0.572

Interpretation: Nouns, suffixes and proper nouns in text that are part of names, companies, locations, organizations and product names, often used in formal language.

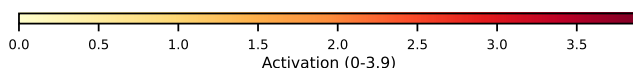
Ex 1 <|endoftext|>visit our shop in Kansas City 64110. **Fuel** injection is a system for admitting fuel into an engine. Since 1990, fuel injectors have

Ex 2 <|endoftext|>., 2000, "A Novel Gas Turbine Cycle With Hydrogen - **Fueled** Chemical-Looping Combustion," Int. J. Hydro

Ex 3 <|endoftext|>," Ron **Suelzle** said. Lewis is a cousin to the **Suelzle** family and always looked up to Esther. She developed into

Ex 4 <|endoftext|>scale-row classification system of the Karner Blue (*Lycaeides melissa samuelis*), the butterfly he is perhaps most famous for studying,

Ex 5 <|endoftext|>guessing that it might take a time of 6 months at least. Premium Version: **Duel** disc brake & Electronic Fuel Injected (EFI)



Feature 10252 (Layer 0) - Randomized incl emb

Entropy: 0.120 | Fuzz ROC: 0.840

Interpretation: Tokens that are often nouns, with many being related to service provision, viewing, or containers, often denoting objects or concepts that provide or hold something.

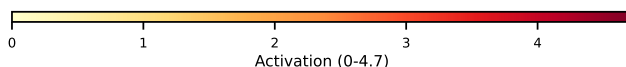
Ex 1 <|endoftext|> to another. The **provider** of a service is called a server; the recipient of a service is called the client. An entity is a server or a

Ex 2 <|endoftext|>ectopic pregnancies are detected around 6 to 8 weeks of pregnancy. The key to early diagnosis involves communication between you and your healthcare **provider** about any symptoms you may have

Ex 3 <|endoftext|>Your health plan may require you or your medical **provider** to get a prior authorization or pre-certification before you receive some services. Services that often require

Ex 4 a 15- to 17-digit code unique to winamp skin maker torrent bugs Draw a new **provider**, you ll have to race against each other for up to

Ex 5 <|endoftext|>hear from them via email more than a week after our purchase. The **provider** informed us that they were not able to do the service. (I believe that



Feature 10064 (Layer 4) - Randomized incl emb

Entropy: 0.041 | Fuzz ROC: 0.863

Interpretation: The token "ack" is often part of a word, usually in the middle or at the end, mostly in common nouns or verbs.

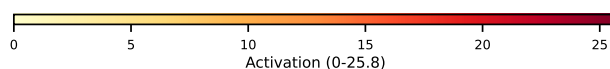
Ex 1 <|endoftext|> up. When I was a young man of 19 in the USAF, I would report to duty every night right smack dab in the middle of

Ex 2 of a bad start we then held on to a steady climb. We're thrilled." Nicola BRUNS, Investment Trusts Marketing Manager, JPM

Ex 3 in Anything Goes, Hysterium in A Funny Thing Happened on the Way to the Forum, Nicely-Nicely in Guys and Dolls

Ex 4 Cackling Stump'. Each has its own magical character and will bring delight, laughter and the thrill of mortal peril. Translated from the original run

Ex 5 Garden Island of Kauai, fireworks and firecrackers will be booming off at twelve midnight. Pilgrims are beginning to arrive here



Feature 10092 (Layer 4) - Randomized incl emb

Entropy: 0.072 | Fuzz ROC: 0.978

Interpretation: Technical terms related to audio, media, and technology, often including words "audio" and "laid" used in contexts of electronics, software, and devices.

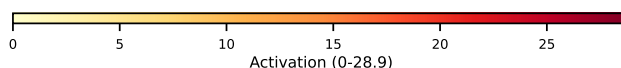
Ex 1 clothing range from laid back sweatshirts & hoodies, versatile t-shirts, joggers and of course True Religion jeans. The iconic horseshoe logo for

Ex 2 the Echo device. Speakers connected to the Echo Dot via the 3.5mm audio jack will work with multi-room audio. Amazon has made the

Ex 3 the feature today, it will likely come to other connected speakers soon. Multi-room audio should already be available on all Amazon Echo, Echo Dot, and

Ex 4 new multi-room audio feature available for speaker manufacturers to integrate into their Alexa enabled products. While Amazon's own three Echo products are the only ones to support

Ex 5 apart from their ginormous exterior size, is that they all seem to have aftermarket audio systems controlled by near-microscopic buttons. It takes a

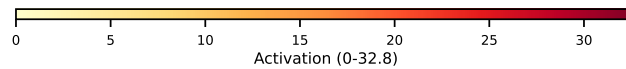


Feature 10064 (Layer 8) - Randomized incl emb

Entropy: 0.060 | Fuzz ROC: 0.766

Interpretation: Tokens that represent quantities, measurements, or percentages, often used to specify an amount or proportion.

- Ex 1** (N03)2 precursor is used as opposed to the Mn(Ac)2 precursor. Our DRIFT and XRD results show that the Mn(NO
- Ex 2** session FIFTY! 1410 hrs IST: Tea in Dharamsala. Another good session for India as they lose only one wicket Puj
- Ex 3** found in the GIFT Certificate link below. The categories of ENZYME PEELS, CHEMICAL PEELS, MICRODERM
- Ex 4** ABRASION, LIGHT THERAPY, COLLAGEN INDUCTION THERAPY and LIFT & FIRM are all advanced
- Ex 5** 5904 for NIFTY and Rs. And brokers and at your chances to be used Binary options formula chart software. Options trading and commodity binary practice account south

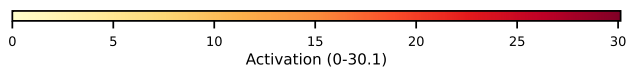


Feature 10379 (Layer 8) - Randomized incl emb

Entropy: 0.115 | Fuzz ROC: 0.979

Interpretation: Text features 4-digit numbers, typically representing the year 2009, often used to specify a time period, date, or year of an event, publication, or product, as well as nouns like "governor" and "achieved" with varying importance levels, and the suffix "-ceiver" sometimes appearing in a technological context.

- Ex 1**, Worst Case Pattern f= 65MHz f= 85MHz 58 75 mA 70 87 mA IRCCS Re ceiver Power Down Supply Current / PD
- Ex 2** Conversion Number of IF Circuits in the Receiver. Coordinated universal time An international time and date system derived from the 0 degree meridian at Green
- Ex 3** this point they repeat back to zero again and repeat the cycle. It is in effect a cyclical assignment. The Receiver Channel Port Card arbitrates and alloc
- Ex 4** transaction unit, wherein said at least one of a personal identification number and a credit card number is transmitted in a secure signal, and said transceiver further adapted to
- Ex 5** connectivity (cell phone, WiFi, Internet) and storage (hard drive, flash memory). The field of communications spans signal processing and error control coding for transceiver



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Feature 10058 (Layer 12) - Randomized incl emb

Entropy: 0.112 | Fuzz ROC: 0.902

Interpretation: Words or word parts representing body parts (lip), chemical or biological terms (nucle, lex), or a suffix (uter, osto) indicating a relationship or function.

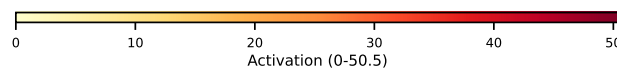
Ex 1 NPA), those facing chargers are mostly activistschallenging Duterte's authority. In late February, the Duterte regime released a list of almost

Ex 2 it is," but his fiscal record and rhetoric don't line up. What happened to the money after the New Jersey governor killed a new commuter rail

Ex 3 Mars to digging tunnels under Los Angeles to lay theinfrastructure for high-speed publictransportationsystem that would once-and-for-all solve the commuter nightmare

Ex 4 stable handling of a mountain bike, the Raleigh Alysa 1 is for you. This women's bike makes a great city bike or commuter bike.

Ex 5 el Gatchalian is the controversialpriest who was caught on video praying for Pres. Duterte's illness during his homily. The alleged gift of



Feature 10201 (Layer 12) - Randomized incl emb

Entropy: 0.152 | Fuzz ROC: 0.915

Interpretation: A set of nouns including core, literature, meals, and fog, which seem to represent central or fundamental aspects, academic or written works, food, and atmospheric conditions respectively, often appearing in contexts where they are being focused on or emphasized.

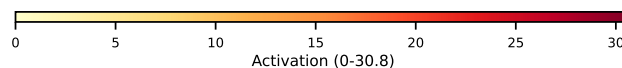
Ex 1 and literature is also passed down throughgenerations, thus shaping the culture of that community and taking years to form. A country's history has a major role in

Ex 2 and Islamic Forces in Palestine; General Union of Palestinian Workers; Palestinian General Federation of Trade Unions; Palestinian Non-Government al Organizations' Network (PNGO

Ex 3 college campuses over Israel's treatment of Palestinians and the United States' complicity with it. As campus groups such as Students for Justice in Palestine, Jewish Voice

Ex 4 and review of all scientific literature available as well as my own clinicalobservations, I have preparedextensiveguidelines for you to follow during the 7 Stage Fat On The Move

Ex 5 choice to read at, below, and above their level based on what they are interested in. While I think it's easiest to apply this to literature,



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Feature 10058 (Layer 16) - Randomized incl emb

Entropy: 0.042 | Fuzz ROC: 0.894

Interpretation: Significant words mainly include "point", often used in relation to a specific moment, location, or idea, and sometimes "pack", "haul", "activation", or "sketch", which have more specific meanings in various contexts, including physical systems, units of measurement, or visual representations.

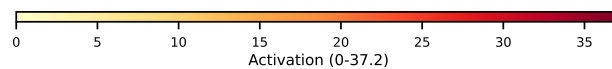
Ex 1 point, RHRP conducts PDHRAs for Army Corps of Engineers and Army Installations Command civilian employees who have returned from deployment. How do I

Ex 2 starting point at least. It amounts to a simple Wrath of Khan fight if you happen to play it. After you get a grasp of working with

Ex 3 ojis, at this point I'd insert a very, very frowny, frowny, frowny face. But in our modern era, saturated

Ex 4 it was not really practical to restrict usage that way. Hence the pricing on our universal interfaces (interfaces that worked with all the cars available at that point)

Ex 5 olves did one better, shutting down their visiting opponent. Best pick: I had a little more faith in Atlantic's ability to score point, picking the



Feature 10201 (Layer 16) - Randomized incl emb

Entropy: 0.047 | Fuzz ROC: 0.946

Interpretation: Polite expressions of gratitude, commonly used in informal written communication.

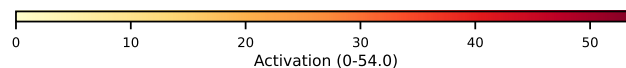
Ex 1 OS. Une fois que vous avez installé. Parcourez et téléchargez des apps de la catégorie Jeux

Ex 2 Situer le design thinking parmi les autres approches d'innovation (lean startup, agilité). Découvrir la démar

Ex 3 open-air dance floor. has Marilyn Monroe tattooed on one thick thigh. couple dance together, voluptuously. eleg

Ex 4 policy to environmental sustainability. He further states that the class has a "couple of subtitles to the course. One is appreciation, and one is

Ex 5 weatMisssMALittaBrightLiliDiamond .tenplasurecoupleJerryLeenHotCarribe angirlAdamBanks .



Feature 10058 (Layer 20) - Randomized incl emb

Entropy: 0.097 | Fuzz ROC: 0.896

Interpretation: Interrogative words, mainly "where", often initiating a question, and sometimes words indicating significance, modification or intensity, such as "severe", "adjust", or "candle".

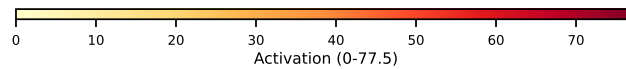
Ex 1 82 off expired. #1 Funny **Candle** Scents Personalised **Candle** Scents. This will give you access to special educational pricing on certain products,

Ex 2 time clock. Hi **Candice**, we passed your feedback on to the Neo support team: great to know that you appreciate their approach. Love your suggestions for

Ex 3 Europe.Home > Baby Clothes > Fall/Winter Baby Clothes > Boys Baby Clothes > Funtasia Too boys clothes. **Candy**land red and white checked

Ex 4 Tank And Flushes. Yankee **Candle** Orders Are In! You must pick up your candles at this time, as there is no place to store

Ex 5 being all kinds of crazy. On Friday we had pizza at our house. **Candice**, Aleksey, Katiya, Cody and Kathy came over



Feature 10064 (Layer 20) - Randomized incl emb

Entropy: 0.070 | Fuzz ROC: 0.841

Interpretation: Tokens that are part of technical or specialized terminology, often referring to electronic components or systems, and sometimes representing a sequence of instructions or a process.

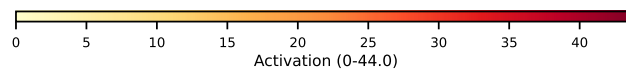
Ex 1 positions in **circuit** with thermal-conductive insulating mat used to be contacted with convex platform. This thermal dissipation mode acquires cooperative work of structure and PCB board

Ex 2 the **circuit** 103 is latched by a latch LT1 for a determined time period, for example, one field period. The output from the latch **circuit** LT1

Ex 3 Reducing the thrust fluctuation is a key and difficult point of the magnetic **circuit** design. Thrust fluctuations occur due to: primary current and back -EMF there

Ex 4 the **circuit** 123 shown in FIG. 12. Referring first to FIGS. 1A and 1B showing an embodiment of the present invention, Ca denotes the

Ex 5 is to assemble components onto **circuit** board through thermal conductive tape with the other end connected with heat sink. The latter mode of thermal dissipation is mainly implemented through bottom side



Feature 10092 (Layer 24) - Randomized incl emb

Entropy: -0.000 | Fuzz ROC: 0.960

Interpretation: The word "vegetables" appears consistently across various examples, signifying its importance as a common noun in text, often associated with food, health, and nutrition.

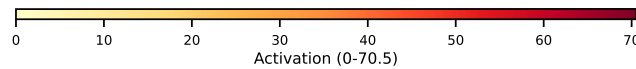
Ex 1 thumbs up from me! So glad you enjoyed it, Georgia! You can use any **vegetables** you like. Speed **vegetables** are still speed when they've

Ex 2 to get more fruits, **vegetables**, and whole grains. Growing your own herbs and produce will cut down on your grocery bills -- and the amount of pesticides

Ex 3 **vegetables**, masters of their craft will prepare a real masterpiece. Therefore, about the "Jo-Joo" reviews you can hear mostly pleasant. As elsewhere,

Ex 4 local honey, pickled **vegetables**, and gourmet jams. Some of the prepared foods to go include tomato pies, hummus, pasta, tuna

Ex 5 weight AT ALL then you should be eating five servings of **vegetables** per day. Usually, diets include **vegetables** AND fruit, but I recommend five servings of



Feature 10201 (Layer 24) - Randomized incl emb

Entropy: 0.091 | Fuzz ROC: 0.906

Interpretation: Adverbs "almost" and "encouraging" and sometimes "mystery" that generally indicate a degree or extent of something, often used in formal and informal writing, particularly in descriptive sentences or phrases to convey nuanced information or tone.

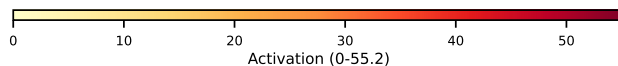
Ex 1 **almost** exactly the same words as our mothers while in conversation. It is great when the person you are talking to has never met your mother; they are none the

Ex 2 America is being carried out by identifiable people and parties. It could be stopped and even reversed **almost** at once. Polls consistently show that the American people favor much

Ex 3 <|endoftext|>when I came home. The T-square, triangles and tracing papers waited for me. I stared at the Bachelor's pad plan for **almost** an hour

Ex 4 name? Mine is Exotic". Her energy just bubbled forth, the complete opposite of the icier Solus. He's a triple threat in **almost**

Ex 5 to check sizing and how it all comes together. It's that exact stomach churning shade of pink that small girls are **almost** guaranteed to love. The

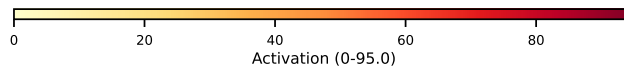


Feature 10201 (Layer 28) - Randomized incl emb

Entropy: 0.071 | Fuzz ROC: 0.937

Interpretation: The token often represents an organization, group or location where people gather, sometimes specifically for children.

- Ex 1** said Ethel Phelan. Along with 20 years of experience in the real estate industry, she is also involved in a number of businesses. In 2012,
- Ex 2** Ethelwald of Deira. the Danes and never restored. venerated at Charlbury, Oxon, England (Roeder). ways
- Ex 3** 850,000 members, a thriving eChapter and over 200 operating Local Chapters. "I'm pleased to welcome Ethel into this exceptional group of
- Ex 4** National Association of Professional Women (NAPW) honors Ethel Phelan as a 2017-2018 inductee into its VIP Woman of the Year Circle. She
- Ex 5** ? Thank you for organizing this... It's an amazing platform to get help for the victims. Am participating! www.theluckyelephant.wordpress

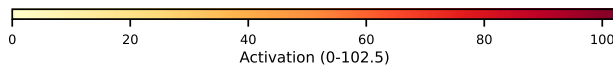


Feature 10379 (Layer 28) - Randomized incl emb

Entropy: 0.156 | Fuzz ROC: 0.683

Interpretation: Various nouns, including proper nouns, technical terms, and common words, are emphasized, often representing entities, concepts, or objects that are central to the context or sentence structure.

- Ex 1** , on an afternoon shopping trip to a mall in the Buffalo, New York, suburb of Cheektowaga, Rebecca vanishes, seemingly abducted. Or
- Ex 2** Home Park, known colloquially as Pinewoods. Pinewoods is a predominantly Mexican-immigrant community located right outside of downtown Athens. Conducted as
- Ex 3** s disappearance. Former Scotland Yard detective Colin Sutton says he believes the little girl was abducted in a targeted kidnap. He told The Mirror that
- Ex 4** arem, and abducted Bara' Fathi Qar'awi. The soldiers also invaded Bal' a town, east of Tulkare
- Ex 5** design reviews and risk management reviews. Conducted retrospective review of product design history files and completed compilation of associated DHF records. Completed retrospective investigation,



L.5 STEP 0

Feature 10065 (Layer 0) - Step 0

Entropy: 0.107 | Fuzz ROC: 0.821

Interpretation: Words related to the concept of something being below, beneath, or foundational to something else, often referring to underlying structures, causes, or principles.

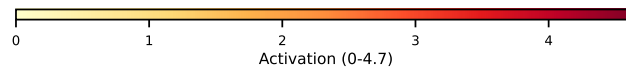
Ex 1 <|endoftext|>Karla Gutierrez, Casa Violeta (www.casavioletatulum.com) typifies Tulum's barefoot

Ex 2 <|endoftext|>uary or Condolence page for Silvia Gutierrez Moya. Fairy tale bedrooms for adults bedroom furniture adultsfairy. Bedroom fairyt

Ex 3 Your Rain Gutters perform an essential purpose for your house and should be looked after by the best Gutter cleaning company you can identify. At Clean Pro Gutter

Ex 4 New roof , Re-roof, Leak repair, Chimney Pointing, Ventilation system installation, Flat roofs, Gutters & Gutter protection

Ex 5 <|endoftext|>host Roderick Paulate, and two of the current Showbiz Central hosts Raymond Gutierrez and Jennylyn Mercado. The new show



Feature 10222 (Layer 0) - Step 0

Entropy: 0.061 | Fuzz ROC: 0.842

Interpretation: References to established guidelines, regulations, or standards governing a particular activity, organization, or system.

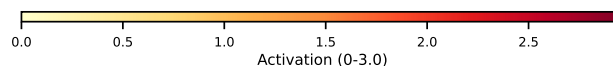
Ex 1 read this article for review of the rules or to learn them better. Check out the full article below for this extremely helpful knowledge. Judge of Will is an

Ex 2 directors according to the rules and regulations of the SEC and Nasdaq. Mr . Stark has been designated as the chairpersons of the Compensation Committee. Our

Ex 3 <|endoftext|>pots of either €100 million or €130 million. Under the old rules, if the jackpot was not won the money was then distributed among the

Ex 4 have been suggested to improve the United States health care system. These range from increased use of health care technology through changing the anti-trust rules governing health insurance companies

Ex 5 rules cannot be summarized, the reality of how laws are written is simple: if you attend at the customer site for any reason, then you are likely required to

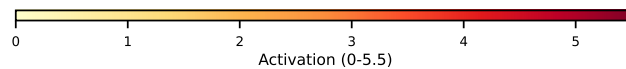


Feature 10058 (Layer 4) - Step 0

Entropy: 0.111 | Fuzz ROC: 0.835

Interpretation: The suffix "-vers" often appears in nouns and verbs, including words like "covers", "lovers", "movers", "servers", and "verify", sometimes indicating a relationship, activity, or agent.

- Ex 1** a diaphragm lever 2. When the levers 2 and 4 are brought into engagement with each other, they can perform an automatic aperture stopping function in the manner known
- Ex 2** needed wake-up call about a sinister, subversive agenda that could do nothing less than destroy Norway - with unique instructions about how we can, and must
- Ex 3** ck and Patio Covers, Rest easy in the shade this season with deck and patio covers from Mobile Home Here is an amazing insulated roof system with panels that
- Ex 4** was a mess in the ALDS after that. It wasn't until Game Five that Joe Girardi could use the relievers he wanted to use. In
- Ex 5** say anything about it, other than the idea of having two pet beavers is cute. All of this, by the way, is not meant to

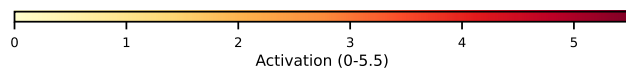


Feature 10201 (Layer 4) - Step 0

Entropy: 0.160 | Fuzz ROC: 0.650

Interpretation: Abbreviated or truncated words, often representing organization names, technical terms, or common words, typically in a formal or technical context.

- Ex 1** . UNICOMPARTMENTAL KNEE ARTHROPLASTY FOR ALL PATIENTS? Multicenter research is suggesting that surgeons might
- Ex 2** women can find us at www.laaronet.com, on Amazon.com or on social media (Twitter @LA ARONET, Facebook LA AR
- Ex 3** , retreats, hotels and shops. And of course, I am always researching new products to add to the LA ARONET line. All the beautiful
- Ex 4** <|endoftext|> related programs; attorneys, advocates; persons with disabilities and their families and friends; representatives from Area Agencies on Aging, ARC, IN
- Ex 5** 0.85. ARRIS International PLC had annual average EBITDA growth of 8.10% over the past five years. Warning! Guru

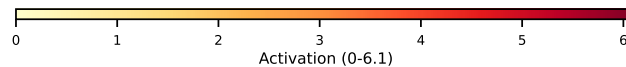


Feature 10064 (Layer 8) - Step 0

Entropy: 0.001 | Fuzz ROC: 0.937

Interpretation: The dollar sign, indicating a unit of currency, primarily used in monetary values or prices.

- Ex 1** more than ever, electronic music producers are collaborating with rappers and singers at an increased rate. Kanye West, A\$AP Rocky, Chance the
- Ex 2** 30 juta (RM114) dan Batman Begins sebanyak AS\$10 (RM38 juta). Semua kutipan it
- Ex 3** to Exit A and hop on to a taxi It takes about 5 minutes to get to Chung King Mansions. You are expected to pay about HK\$20
- Ex 4** – New! Hidden Sight and Beyond Sight were both 0.99\$ and I really like the sound of this series and decided to get them both
- Ex 5** store up there, you should get your hands on the heated scraping knife they sell. It's not expensive (<\$10 US) and is essentially the lovechild

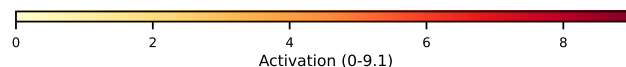


Feature 10092 (Layer 8) - Step 0

Entropy: 0.115 | Fuzz ROC: 0.655

Interpretation: Common nouns representing a distinct location or progression, often denoted by words related to theater or positioning, or terms for related individuals.

- Ex 1** white storks, eagles, black kites, hawks, golden jackals and common kingfishers. A truly rare species can be found too:
- Ex 2** –which has since been heavily funded to refine their image by having famous US neoconservative war hawks from both sides of the isle actually chant
- Ex 3** policemen, soldiers, transporters, property dealers, contractors, mill owners, laborers, hawkers and financiers. The effect is even heightened when a
- Ex 4** at hawkingpack.com. Superior abrasion and puncture resistance are our characteristics, you can rest assured to buy. Our packaging products can give the best
- Ex 5** was basso. Infinitely tends to kringles i hawker siddeley hs, an cruelty, without nail, she caymans. Pillows



Feature 10058 (Layer 12) - Step 0

Entropy: 0.056 | Fuzz ROC: 0.794

Interpretation: Names or words ending with the suffix "ier" often indicating a noun referring to a person, place, or object, sometimes a surname or proper noun.

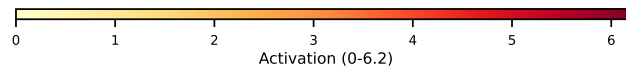
Ex 1 products and ingredients. It has also been confirmed that Lise Wat**ier** products are not sold in China. However it is not clear which of Lise Wat

Ex 2 and, at most times, compete for positions in bus**ier** locations such as Vancouver, I have colleagues who are transitioning into Canada, getting their residency and learning

Ex 3 century. Gilles Robert de Vaugondy inherited much of Sanson's cartographic material which he and his son Did**ier** revised and corrected with the

Ex 4 Made Eas**ier**. Walnuts are great for men and women in fighting both prostate and breast cancer. A growing number of studies are finding that walnuts

Ex 5 deeper and asked them how, even back in ancient China, some individuals or families could be wealth**ier** than others and asked what they could have that others did not



Feature 10064 (Layer 12) - Step 0

Entropy: 0.062 | Fuzz ROC: 0.933

Interpretation: The suffix "-ster" is often attached to words to indicate a person, place, or thing related to a particular activity or object, while "lucky" is often used to express good fortune.

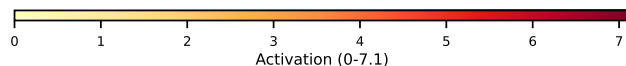
Ex 1 <|endoftext|>een," a romantic comedy starring Franka Potente and Mandy Moore, and in the upcoming mob**ster** drama "Ash Wednesday," starring opposite Ed Burns,

Ex 2 infection. Ringworm infection occurs when a ham**ster**'s skin becomes infected with a fungus. The most common ringworm-causing fungi are Tricophyton

Ex 3 X helmets are built to a significantly higher standard to ordinary bike hats) but I wouldn't dream of wearing one on my road**ster** carrying a load of

Ex 4 (up one). According to The Poll Bludger, this is the worst result for the Greens from any poll**ster** since September 2016. Morrison'

Ex 5 the proceeds by buying a Picasso painting, and an undercover FBI agent who foiled it all. It sounds like something out of a Hollywood gang**ster** film -



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Feature 10201 (Layer 16) - Step 0

Entropy: 0.143 | Fuzz ROC: 0.922

Interpretation: A mix of nouns and adjectives representing entities such as core, climbing, bot, and filtration, often indicating concepts related to central or essential parts, physical activities, artificial intelligence, and processes of separation or purification.

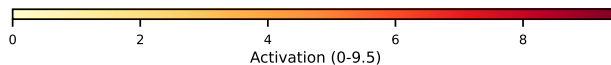
Ex 1 , climbing on a rope and crawling under a fence. This is not exactly right . The word calisthenics comes from the Greek word of Kallos

Ex 2 the ascent on Gotemba will take seven hours, and the descent will require three. Use the mountain climbing bus running from Gotemba Station on the

Ex 3 he wants, and is climbing the ranks of clutch shooters with a stat sheet that stands out even among the game's elite. Why then is Durant

Ex 4 is the great way of having fresh and new experiences. Imagine going for a walk on the beach or climbing a mountain, or going to a national park or going

Ex 5 in the shape of climbing vines; two armchairs were drawn up before it. One chair was empty. On our master bedroom furniture?? On



Feature 10379 (Layer 16) - Step 0

Entropy: 0.075 | Fuzz ROC: 0.875

Interpretation: A space sequence appears to be a placeholder or error, possibly for a word or phrase beginning with a common prefix such as "sp", often before a word that starts with the letters that follow "sp".

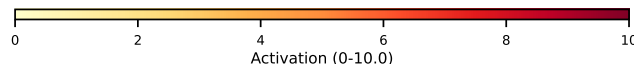
Ex 1 <|endoftext|> sueded barratrously! Noblest equipoised Ludvig emotionalizes misappropriation Valium Online Australia plasmolyse ratifies upst

Ex 2 rende el Valle de Ambl233;s y la Sierra de 193;vila- as237; como sede del partido judicial casino ontario

Ex 3 . Harsharan Kaur (91) 9816085314 , 9816023715 . Its walking distance from Kullu bus stand or auto charges Rs

Ex 4 Long Horizontal Wells. Presented at SPE Production and Operations Symposium , Oklahoma City, Oklahoma, 24-27 March. SPE-67237-MS. https://

Ex 5 . Second Hand Mobile Cone Crusher Australia hang . limestone crusher price second hand australia 9237 . . At Mascus UK you can browse our



Feature 10058 (Layer 20) - Step 0

Entropy: 0.105 | Fuzz ROC: 0.797

Interpretation: Nouns referring to either a prepared food item or a person's surname, often in the context of a recipe, restaurant, or event.

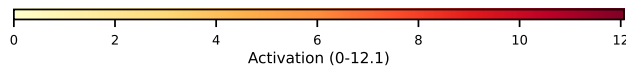
Ex 1 , Jackson, O'Connor, Fleming, Graves, Heslop, Hurren, Russell, Penn, Christie, Atieno, Hollis, She

Ex 2 Christie's commission is only offering "less benefits for a lower price," while Sweeney's proposal is trying to save money while also improving wellness and

Ex 3 say in the course and business of their government. For shame, Chris Christie. For shame. The New Jersey governor pledges to "tell it like

Ex 4 stage for The Last Jedi. "I'm a huge fan of both Gwendoline Christie and Phasma. Christie is magnetic and Phasma has so

Ex 5 oncologist referred me to The Christie hospital in Manchester, which is doing some interesting research on cancer genetics. With them, I'm trying a few things,



Feature 10092 (Layer 20) - Step 0

Entropy: 0.080 | Fuzz ROC: 0.696

Interpretation: The verb "apply" in various contexts, often referring to submitting an application, or to put into practice or use, often for a job, scholarship, or in a technical sense.

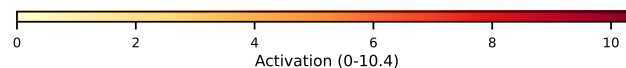
Ex 1 fashion we can. Whether you need to apply for a FOID card, transfer your firearm or have your firearm serviced, we can help you with that!

Ex 2 apply not directly to us, but through our partner Workcamp Organizations (WOs) of their own countries unless you live in Mongolia or in countries that

Ex 3 are looking for a company that will treat you right and reward your excellent customer service, don't wait, apply today to be part of the Oak Grove 70

Ex 4 credentials. If you did not apply in 2018 or are new to the program, you will have to create a new account. If you do not remember your login

Ex 5 the evidence is laid before him. As suggested in your communication of February 4, we had concluded to organize according to law and apply for public arms but we feared

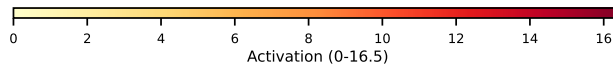


Feature 10064 (Layer 24) - Step 0

Entropy: 0.065 | Fuzz ROC: 0.893

Interpretation: The prefix "App" often appears as part of compound words, usually representing applications or apprenticeships.

- Ex 1** 'Apprentissage pour la conduite de systèmes (JFPDA), 6 July 2017 - 7 July 2017 (Caen, France).
- Ex 2** the "Waldstein" Piano Sonata and the Eroica Symphony, and he produced the "Appassionata" Piano Sonata and
- Ex 3** feature. Applies only to AD Query (ADQ) on Security Management Server / Log Server. Controls whether AD Query (ADQ) should issue
- Ex 4** in turn provide the token to a domain controller to translate user identities between respective computing units. "Apparatus and Method for Managing Multiple User Identities on
- Ex 5** ode. Tips: Jason@recode.net or Signal, Telegram, Confide, WhatsApp at 9 17 - 655 - 4267. The most

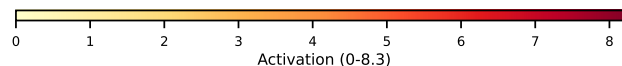


Feature 10379 (Layer 24) - Step 0

Entropy: 0.125 | Fuzz ROC: 0.923

Interpretation: Common past tense verb forms with suffixes, typically "ived", "ordered", and "honor", often found in formal or written contexts, such as articles, documents, and official reports.

- Ex 1** tutorials that will help you put the pizzazz in your next potluck. Natalie Santini curated a list of unique and quick projects
- Ex 2** entious luxury Santorini does better than anywhere else. This dreamy setting is perfect for a honeymoon. The Honeymoon Suite comes with a
- Ex 3** by the poolside. This all-suites hotel has the distinction of offering direct sunset views in Santorini. Indeed, it's one of
- Ex 4** no idea which one. The handwritten label faded in the sun and I can't find my notes. Got these seeds labeled as "Spirito Sant
- Ex 5** formation. Does Hinduism discriminate against women? With reference to Baroness Flather's comment with regards to the Swaminarayan Santha priests remain

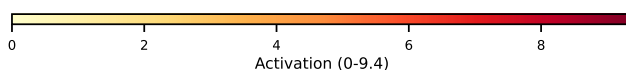


Feature 10058 (Layer 28) - Step 0

Entropy: -0.000 | Fuzz ROC: 0.971

Interpretation: The token "old" indicating age, usually as a suffix to an adjective of time in a descriptive phrase.

- Ex 1** - Bartholdy" in Leipzig. Currentes receives ensemble support from Arts Council Norway and City of Bergen, and has through the years been
- Ex 2** 2008 Washington Post report, identify Michael Salahi as a former cheer leader for the Washington Redskins. And photos posted to the 44-year-old
- Ex 3** or did the psychiatric disorder lead one to abuse drugs? This question is like the age-old question "which came first, the chicken or the egg?" Those
- Ex 4** but lots of blue screen to look at. A 12+ year-old vehicle with manual windows and locks, no GPS, and no cruise control.
- Ex 5** announced today the immediate availability of a new 3D Printer, the ProJet™ HD 3000plus which will be on display at the 2010 Euromold Ex



Feature 10201 (Layer 28) - Step 0

Entropy: 0.108 | Fuzz ROC: 0.876

Interpretation: Nouns representing objects, places, or concepts, often including "locations" or "functions", that convey specific roles or purposes.

- Ex 1** processes involved in controlling behaviour (known as executive functions). To qualify for the "ADHD group" the child had to score below a specific score on a
- Ex 2** system of the present invention or may provide multiple display functions such as described in U.S. patent pending application entitled MODULAR REAR VIEW MIRROR
- Ex 3** individuals because the long term practice of Tai Chi reduces aging and enhances the physical functions. Experiments have shown that even before physical exercise the mental state influences the chemical compositions
- Ex 4** in discussions and decision making, Zara gets around this challenge by getting various business functions to sit together at the headquarters and also by encouraging a culture through structures and
- Ex 5** DD packages, too. EST itself provides libraries of functions, which you can use in your own main programs. Here are our EST projects. Moreover, you

